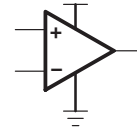


TLV4110, TLV4111, TLV4112, TLV4113 FAMILY OF HIGH OUTPUT DRIVE OPERATIONAL AMPLIFIERS WITH SHUTDOWN

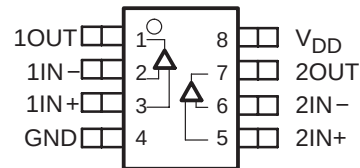
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- High Output Drive . . . >300 mA
- Rail-To-Rail Output
- Unity-Gain Bandwidth . . . 2.7 MHz
- Slew Rate . . . 1.5 V/ μ s
- Supply Current . . . 700 μ A/Per Channel
- Supply Voltage Range . . . 2.5 V to 6 V
- Specified Temperature Range:
 - $T_A = 0^\circ\text{C}$ to 70°C . . . Commercial Grade
 - $T_A = -40^\circ\text{C}$ to 125°C . . . Industrial Grade
- Universal OpAmp EVM

Operational Amplifier



TLV4112
D, DGN, OR P PACKAGE
(TOP VIEW)



description

The TLV411x single supply operational amplifiers provide output currents in excess of 300 mA at 5 V. This enables standard pin-out amplifiers to be used as high current buffers or in coil driver applications. The TLV4110 and TLV4113 come with a shutdown feature.

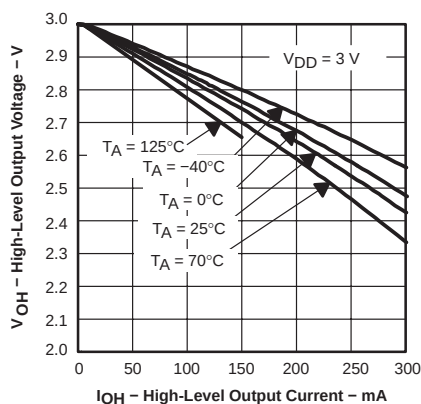
The TLV411x is available in the ultra small MSOP PowerPAD™ package, which offers the exceptional thermal impedance required for amplifiers delivering high current levels.

All TLV411x devices are offered in PDIP, SOIC (single and dual) and MSOP PowerPAD (dual).

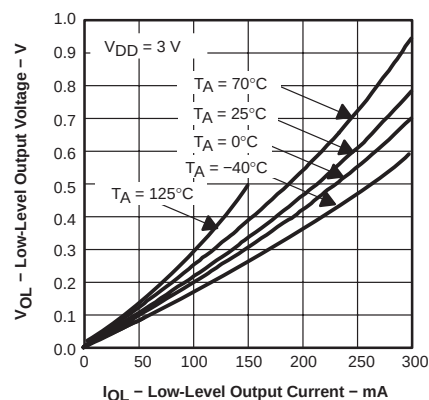
FAMILY PACKAGE TABLE

DEVICE	NUMBER OF CHANNELS	PACKAGE TYPES			SHUTDOWN	UNIVERSAL EVM BOARD
		MSOP	PDIP	SOIC		
TLV4110	1	8	8	8	Yes	Refer to the EVM Selection Guide (Lit# SLOU060)
TLV4111	1	8	8	8	—	
TLV4112	2	8	8	8	—	
TLV4113	2	10	14	14	Yes	

HIGH-LEVEL OUTPUT VOLTAGE
vs
HIGH-LEVEL OUTPUT CURRENT



LOW-LEVEL OUTPUT VOLTAGE
vs
LOW-LEVEL OUTPUT CURRENT



Please be aware that an important notice concerning availability, standard warranty, and use in critical applications of Texas Instruments semiconductor products and disclaimers thereto appears at the end of this data sheet.

PowerPAD is a trademark of Texas Instruments. All other trademarks are the property of their respective owners.

PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.



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TLV4110, TLV4111, TLV4112, TLV4113

FAMILY OF HIGH OUTPUT DRIVE OPERATIONAL AMPLIFIERS WITH SHUTDOWN

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TLV4110 AND TLV4111 AVAILABLE OPTIONS

T _A	PACKAGED DEVICES			
	SMALL OUTLINE (D) ^{†‡}	MSOP		PLASTIC DIP (P)
		SMALL OUTLINE (DGN) [†]	SYMBOL	
0°C to 70°C	TLV4110CD	TLV4110CDGN	xxTIAHL	TLV4110CP
	TLV4111CD	TLV4111CDGN	xxTIAHN	TLV4111CP
–40°C to 125°C	TLV4110ID	TLV4110IDGN	xxTIAHM	TLV4110IP
	TLV4111ID	TLV4111IDGN	xxTIAHO	TLV4111IP

[†] This package is available taped and reeled. To order this packaging option, add an R suffix to the part number (e.g., TLV4110CDR).

[‡] In the SOIC package, the maximum RMS output power is thermally limited to 350 mW; 700 mW peaks can be driven, as long as the RMS value is less than 350 mW.

TLV4112 AND TLV4113 AVAILABLE OPTIONS

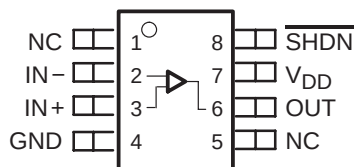
T _A	PACKAGED DEVICES					
	SMALL OUTLINE (D) ^{†‡}	MSOP				PLASTIC DIP (P)
		SMALL OUTLINE (DGN) [†]	SYMBOL	SMALL OUTLINE (DGQ) [†]	SYMBOL	
0°C to 70°C	TLV4112CD	TLV4112DGN	xxTIAHP	—	—	TLV4112CP
	TLV4113CD	—	—	TLV4113CDGQ	xxTIAHR	TLV4113CN
–40°C to 125°C	TLV4112ID	TLV4112IDGN	xxTIAHQ	—	—	TLV4112IP
	TLV4113ID	—	—	TLV4113IDGQ	xxTIAHS	TLV4113IN

[†] This package is available taped and reeled. To order this packaging option, add an R suffix to the part number (e.g., TLV4112CDR).

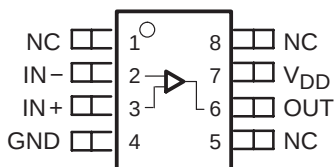
[‡] In the SOIC package, the maximum RMS output power is thermally limited to 350 mW; 700 mW peaks can be driven, as long as the RMS value is less than 350 mW.

TLV411x PACKAGE PIN OUTS

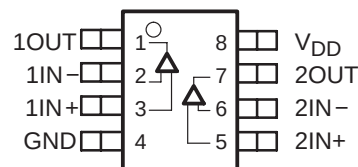
TLV4110
D, DGN OR P PACKAGE
(TOP VIEW)



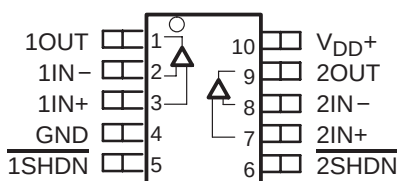
TLV4111
D, DGN OR P PACKAGE
(TOP VIEW)



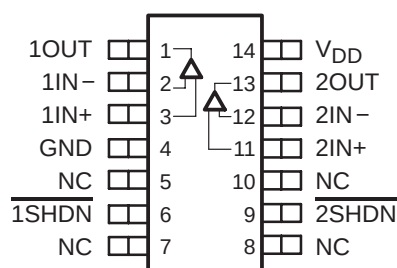
TLV4112
D, DGN, OR P PACKAGE
(TOP VIEW)



TLV4113
DGQ PACKAGE
(TOP VIEW)



TLV4113
D OR N PACKAGE
(TOP VIEW)



NC – No internal connection

TLV4110, TLV4111, TLV4112, TLV4113 FAMILY OF HIGH OUTPUT DRIVE OPERATIONAL AMPLIFIERS WITH SHUTDOWN

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absolute maximum ratings over operating free-air temperature range (unless otherwise noted)[†]

Supply voltage, V_{DD} (see Note 1)	7 V
Differential input voltage, V_{ID}	$\pm V_{DD}$
Input voltage range, V_I	$\pm V_{DD}$
Output current, I_O (see Note 2)	800 mA
Continuous /RMS output current, I_O (each output of amplifier): $T_J \leq 105^\circ\text{C}$	350 mA
$T_J \leq 150^\circ\text{C}$	110 mA
Peak output current, I_O (each output of amplifier): $T_J \leq 105^\circ\text{C}$	500 mA
$T_J \leq 150^\circ\text{C}$	155 mA
Continuous total power dissipation	See Dissipation Rating Table
Operating free-air temperature range, T_A : C suffix	0°C to 70°C
I suffix	-40°C to 125°C
Maximum junction temperature, T_J	150°C
Storage temperature range, T_{stg}	-65°C to 150°C
Lead temperature 1,6 mm (1/16 inch) from case for 10 seconds	260°C

[†] Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

- NOTES: 1. All voltage values, except differential voltages, are with respect to GND.
2. To prevent permanent damage the die temperature must not exceed the maximum junction temperature.

DISSIPATION RATING TABLE

PACKAGE	θ_{JC} ($^\circ\text{C}/\text{W}$)	θ_{JA} ($^\circ\text{C}/\text{W}$)	$T_A \leq 25^\circ\text{C}$ POWER RATING	$T_A = 125^\circ\text{C}$ POWER RATING
D (8)	38.3	176	710 mW	142 mW
D (14)	26.9	122.3	1022 mW	204.4 mW
DGN (8) [‡]	4.7	52.7	2.37 W	474.4 mW
DGQ (10) [‡]	4.7	52.3	2.39 W	478 mW
P (8)	41	104	1200 mW	240.4 mW
N (14)	32	78	1600 mW	320.5 mW

[‡] See The Texas Instruments document, *PowerPAD Thermally Enhanced Package Application Report* (literature number SLMA002), for more information on the PowerPAD package. The thermal data was measured on a PCB layout based on the information in the section entitled *Texas Instruments Recommended Board for PowerPAD* on page 33 of the before mentioned document.

recommended operating conditions

			MIN	MAX	UNIT
Supply voltage, V_{DD}			2.5	6	V
Common-mode input voltage range, V_{ICR}			0	$V_{DD}-1.5$	V
Operating free-air temperature, T_A	C-suffix		0	70	$^\circ\text{C}$
	I-suffix		-40	125	
Shutdown turn-on/off voltage level [§]	V(on)	$V_{DD} = 3\text{ V}$	2.1		V
		$V_{DD} = 5\text{ V}$	3.8		
	V(off)	$V_{DD} = 3\text{ V}$		0.9	
		$V_{DD} = 5\text{ V}$		1.65	

[§] Relative to GND



TLV4110, TLV4111, TLV4112, TLV4113

FAMILY OF HIGH OUTPUT DRIVE OPERATIONAL AMPLIFIERS WITH SHUTDOWN

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electrical characteristics at recommend operating conditions, $V_{DD} = 3\text{ V}$ and 5 V (unless otherwise noted)

dc performance

PARAMETER	TEST CONDITIONS	$T_A^\dagger$	MIN	TYP	MAX	UNITS
V_{IO} Input offset voltage	$V_{IC} = V_{DD}/2$, $R_L = 100\ \Omega$, $V_O = V_{DD}/2$, $R_S = 50\ \Omega$	25°C		175	3500	μV
		Full range			4000	
αV_{IO} Offset voltage draft		25°C		3		$\mu\text{V}/^\circ\text{C}$
CMRR Common-mode rejection ratio	$V_{DD} = 3\text{ V}$, $R_S = 50\ \Omega$	25°C		63		dB
	$V_{DD} = 5\text{ V}$, $R_S = 50\ \Omega$	25°C		68		
A _{VD} Large-signal differential voltage amplification	$V_{DD} = 3\text{ V}$, $V_{O(PP)} = 0\text{ to }1\text{ V}$	$R_L = 100\ \Omega$	25°C	78	84	dB
			Full range	67		
		$R_L = 10\text{ k}\Omega$	25°C	85	100	
			Full range	75		
	$V_{DD} = 5\text{ V}$, $V_{O(PP)} = 0\text{ to }3\text{ V}$	$R_L = 100\ \Omega$	25°C	88	94	
			Full range	75		
		$R_L = 10\text{ k}\Omega$	25°C	90	110	
			Full range	85		

† Full range is 0°C to 70°C for C suffix and –40°C to 125°C for I suffix. If not specified, full range is –40°C to 125°C.

input characteristics

PARAMETER	TEST CONDITIONS	$T_A^\dagger$	MIN	TYP	MAX	UNITS
I_{IO} Input offset current	$V_{IC} = V_{DD}/2$	25°C		0.3	25	pA
		Full range			50	
					250	
I_{IB} Input bias current	$V_O = V_{DD}/2$, $R_S = 50\ \Omega$	25°C		0.3	50	
		Full range			100	
					500	
$r_{i(d)}$ Differential input resistance		25°C		1000		$\text{G}\Omega$
C_{IC} Common-mode input capacitance	$f = 100\text{ Hz}$	25°C		5		pF

† Full range is 0°C to 70°C for C suffix and –40°C to 125°C for I suffix. If not specified, full range is –40°C to 125°C.



TLV4110, TLV4111, TLV4112, TLV4113 FAMILY OF HIGH OUTPUT DRIVE OPERATIONAL AMPLIFIERS WITH SHUTDOWN

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electrical characteristics at specified free-air temperature, $V_{DD} = 3\text{ V}$ and 5 V (unless otherwise noted) (continued)

output characteristics

PARAMETER		TEST CONDITIONS		T _A [†]	MIN	TYP	MAX	UNITS		
V _{OH}	High-level output voltage	V _{DD} = 3 V, V _{IC} = V _{DD} /2	I _{OH} = −10 mA	25°C	2.7	2.97	V			
				Full range	2.7					
			I _{OH} = −100 mA	25°C	2.6	2.73				
				Full range	2.6					
	V _{DD} = 5 V, V _{IC} = V _{DD} /2	I _{OH} = −10 mA	25°C	4.7	4.96	V				
			Full range	4.7						
		I _{OH} = −100 mA	25°C	4.6	4.76					
			Full range	4.6						
		I _{OH} = −200 mA	25°C	4.45	4.6					
			−40°C to 85°C	4.35						
V _{OL}	Low-level output voltage	V _{DD} = 3 V and 5 V, V _{IC} = V _{DD} /2	I _{OL} = 10 mA	25°C	0.03	0.1	V			
				Full range		0.1				
			I _{OL} = 100 mA	25°C	0.33	0.4				
				Full range		0.55				
	V _{DD} = 5 V, V _{IC} = V _{DD} /2	I _{OL} = 200 mA	25°C	0.38	0.6					
			−40°C to 85°C		0.7					
		I _O	Output current‡	Measured at 0.5 V from rail	V _{DD} = 3 V	25°C		±220		mA
					V _{DD} = 5 V			±320		
I _{OS}	Short-circuit output current‡	Sourcing		25°C	800		mA			
		Sinking			800					

[†] Full range is 0°C to 70°C for C suffix and -40°C to 125°C for I suffix. If not specified, full range is -40°C to 125°C.

[‡] When driving output currents in excess of 200 mA, the MSOP PowerPAD package is required for thermal dissipation.

power supply

PARAMETER	TEST CONDITIONS	T_A	MIN	TYP	MAX	UNITS
I_{DD} Supply current (per channel)	$V_O = V_{DD}/2$	25°C		700	1000	μA
		Full range			1500	
PSRR Power supply rejection ratio ($\Delta V_{DD} / \Delta V_{IO}$)	$V_{DD} = 2.7\text{ to } 3.3\text{ V}, \text{ No load}, V_{IC} = V_{DD}/2\text{ V}$	25°C	70	82		dB
		Full range	65			
	$V_{DD} = 4.5\text{ to } 5.5\text{ V}, \text{ No load}, V_{IC} = V_{DD}/2\text{ V}$	25°C	70	79		
		Full range	65			

[†] Full range is 0°C to 70°C for C suffix and -40°C to 125°C for I suffix. If not specified, full range is -40°C to 125°C.



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electrical characteristics at specified free-air temperature, $V_{DD} = 3\text{ V}$ and 5 V (unless otherwise noted) (continued)

dynamic performance

PARAMETER		TEST CONDITIONS		$T_A^\dagger$	MIN	TYP	MAX	UNITS
GBWP	Gain bandwidth product	$R_L = 100\ \Omega$	$C_L = 10\text{ pF}$	25°C		2.7		MHz
SR	Slew rate at unity gain	$V_{O(pp)} = 2\text{ V}$, $R_L = 100\ \Omega$, $C_L = 10\text{ pF}$	$V_{DD} = 3\text{ V}$	25°C	0.8	1.57		V/ μs
				Full range	0.55			
			$V_{DD} = 5\text{ V}$	25°C	1	1.57		
				Full range	0.7			
ϕ_M	Phase margin	$R_L = 100\ \Omega$,	$C_L = 10\text{ pF}$	25°C		66		
	Gain margin					16		dB
t_s	Settling time	$V(\text{STEP})_{pp} = 1\text{ V}$, $A_V = -1$, $C_L = 10\text{ pF}$, $R_L = 100\ \Omega$	0.1%	25°C		0.7		μs
			0.01%			1.3		

† Full range is 0°C to 70°C for C suffix and -40°C to 125°C for I suffix. If not specified, full range is -40°C to 125°C .

noise/distortion performance

PARAMETER		TEST CONDITIONS		T _A	MIN	TYP	MAX	UNITS
THD+N	Total harmonic distortion plus noise	V _{O(pp)} = V _{DD} /2 V, R _L = 100 Ω, f = 100 Hz	A _V = 1	25°C	0.025			
			A _V = 10		0.035			
			A _V = 100		0.15			
V _n	Equivalent input noise voltage	f = 100 Hz			55			nV/√Hz
		f = 10 kHz			10			
I _n	Equivalent input noise current	f = 1 kHz			0.31			fA/√Hz

shutdown characteristics

PARAMETER		TEST CONDITIONS	$T_A^\dagger$	MIN	TYP	MAX	UNITS
$I_{DD}(\text{SHDN})$	Supply current in shutdown mode (per channel) (TLV4110, TLV4113)	$\text{SHDN} = 0\text{ V}$	25°C		3.4	10	μA
			Full range			15	
$t_{\text{(ON)}}$	Amplifier turn-on time ‡	$R_L = 100\ \Omega$	25°C		1		μs
$t_{\text{(Off)}}$	Amplifier turn-off time ‡				3.3		

† Full range is 0°C to 70°C for C suffix and -40°C to 125°C for I suffix. If not specified, full range is -40°C to 125°C .

‡ Disable time and enable time are defined as the interval between application of the logic signal to SHDN and the point at which the supply current has reached half its final value.

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FAMILY OF HIGH OUTPUT DRIVE OPERATIONAL
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TYPICAL CHARACTERISTICS

Table of Graphs

			FIGURE
V_{IO}	Input offset voltage	vs Common-mode input voltage	1, 2
CMRR	Common-mode rejection ratio	vs Frequency	3
V_{OH}	High-level output voltage	vs High-level output current	4, 6
V_{OL}	Low-level output voltage	vs Low-level output current	5, 7
Z_o	Output impedance	vs Frequency	8
I_{DD}	Supply current	vs Supply voltage	9
k_{SVR}	Power supply voltage rejection ratio	vs Frequency	10
A_{VD}	Differential voltage amplification and phase	vs Frequency	11
	Gain-bandwidth product	vs Supply voltage	12
SR	Slew rate	vs Supply voltage	13
		vs Temperature	14
	Total harmonic distortion+noise	vs Frequency	15
V_n	Equivalent input voltage noise	vs Frequency	16
	Phase margin	vs Capacitive load	17
	Voltage-follower signal pulse response		18, 19
	Inverting large-signal pulse response		20, 21
	Small-signal inverting pulse response		22
	Crosstalk	vs Frequency	23
	Shutdown forward and reverse isolation		24
	Shutdown supply current	vs Free-air temperature	25
	Shutdown supply current/output voltage		26



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TYPICAL CHARACTERISTICS

INPUT OFFSET VOLTAGE
VS
COMMON-MODE INPUT VOLTAGE

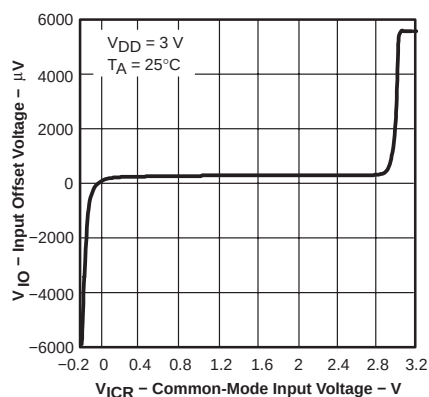


Figure 1

INPUT OFFSET VOLTAGE
VS
COMMON-MODE INPUT VOLTAGE

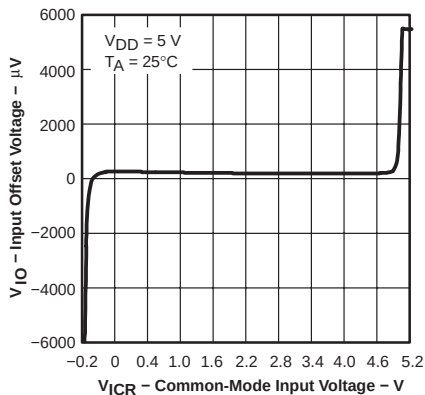


Figure 2

COMMON-MODE REJECTION RATIO
VS
FREQUENCY

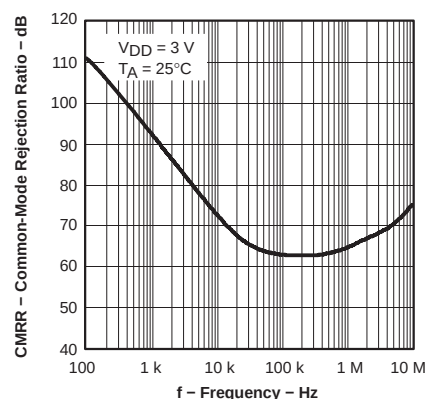


Figure 3

HIGH-LEVEL OUTPUT VOLTAGE
VS
HIGH-LEVEL OUTPUT CURRENT

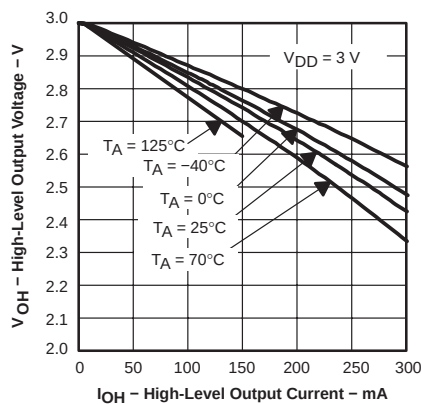


Figure 4

LOW-LEVEL OUTPUT VOLTAGE
VS
LOW-LEVEL OUTPUT CURRENT

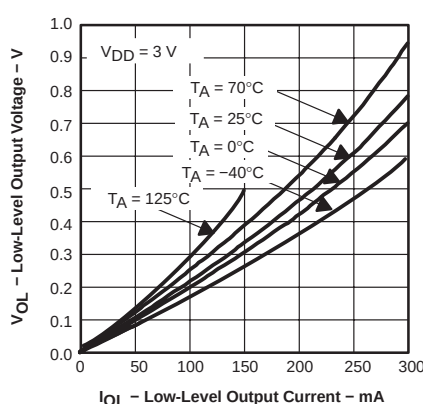


Figure 5

HIGH-LEVEL OUTPUT VOLTAGE
VS
HIGH-LEVEL OUTPUT CURRENT

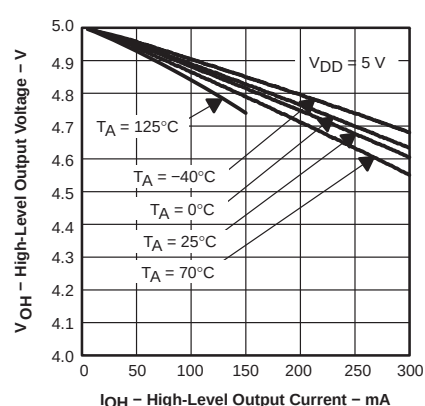


Figure 6

LOW-LEVEL OUTPUT VOLTAGE
VS
LOW-LEVEL OUTPUT CURRENT

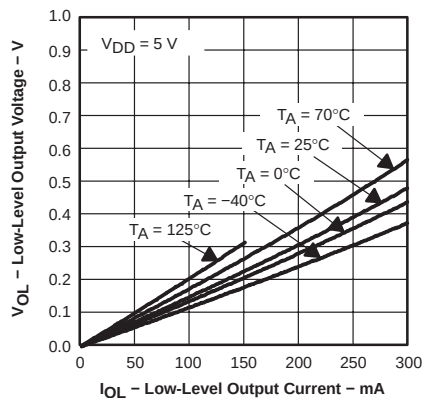


Figure 7

OUTPUT IMPEDANCE
VS
FREQUENCY

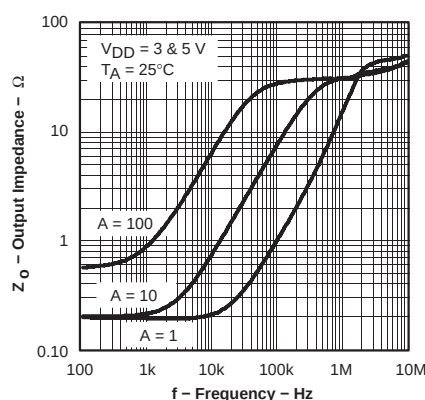


Figure 8

SUPPLY CURRENT
VS
SUPPLY VOLTAGE

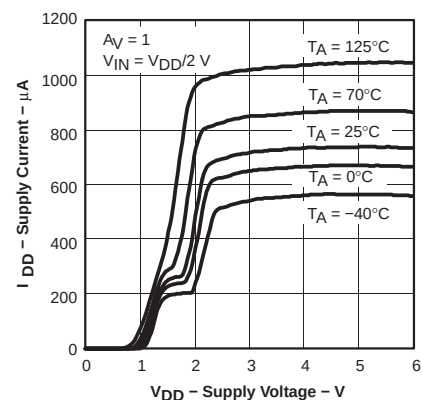


Figure 9

TLV4110, TLV4111, TLV4112, TLV4113 FAMILY OF HIGH OUTPUT DRIVE OPERATIONAL AMPLIFIERS WITH SHUTDOWN

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TYPICAL CHARACTERISTICS

**POWER SUPPLY REJECTION RATIO
VS
FREQUENCY**

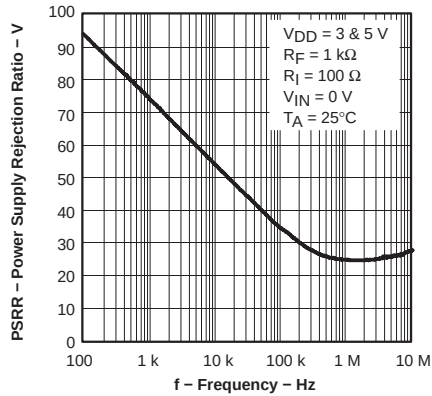


Figure 10

**DIFFERENTIAL VOLTAGE
AMPLIFICATION AND PHASE
VS
FREQUENCY**

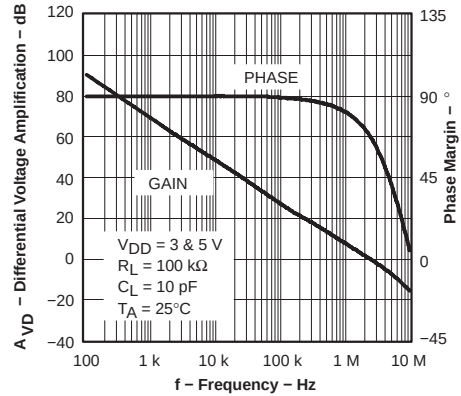


Figure 11

**GAIN-BANDWIDTH PRODUCT
VS
SUPPLY VOLTAGE**

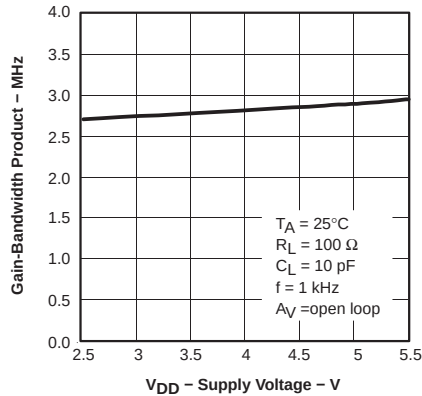


Figure 12

**SLEW RATE
VS
SUPPLY VOLTAGE**

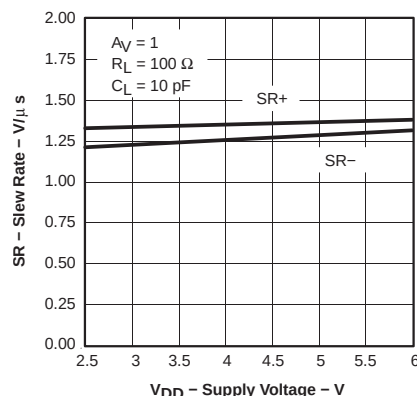


Figure 13

**SLEW RATE
VS
TEMPERATURE**

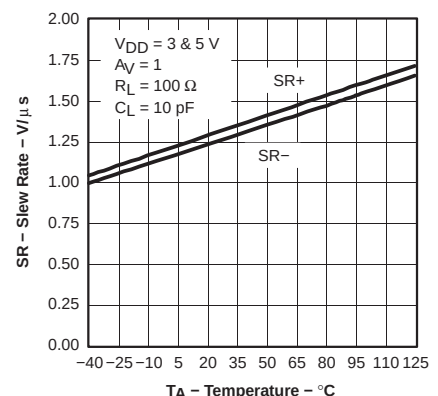


Figure 14

**TOTAL HARMONIC DISTORTION+NOISE
VS
FREQUENCY**

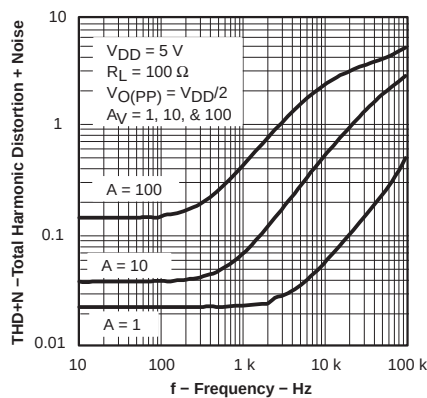


Figure 15

**EQUIVALENT INPUT VOLTAGE NOISE
VS
FREQUENCY**

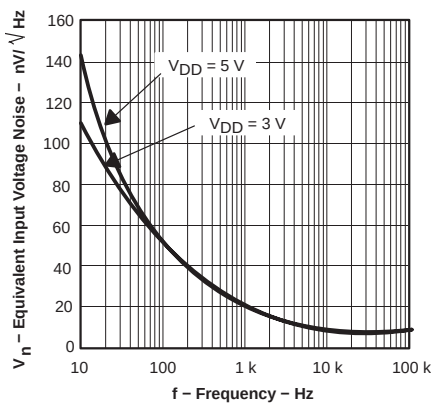


Figure 16

**PHASE MARGIN
VS
CAPACITIVE LOAD**

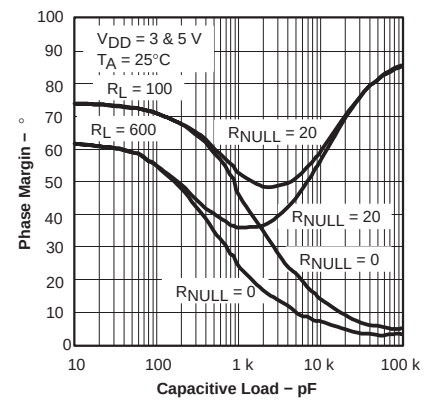


Figure 17

TLV4110, TLV4111, TLV4112, TLV4113

FAMILY OF HIGH OUTPUT DRIVE OPERATIONAL AMPLIFIERS WITH SHUTDOWN

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TYPICAL CHARACTERISTICS

**VOLTAGE-FOLLOWER
LARGE-SIGNAL PULSE RESPONSE**

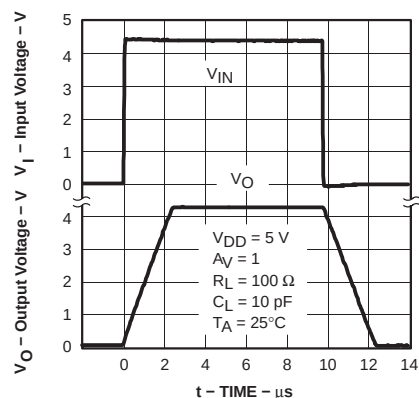


Figure 18

**VOLTAGE-FOLLOWER
SMALL-SIGNAL PULSE RESPONSE**

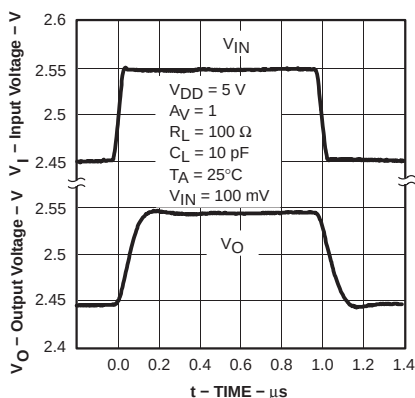


Figure 19

**INVERTING LARGE-SIGNAL
PULSE RESPONSE**

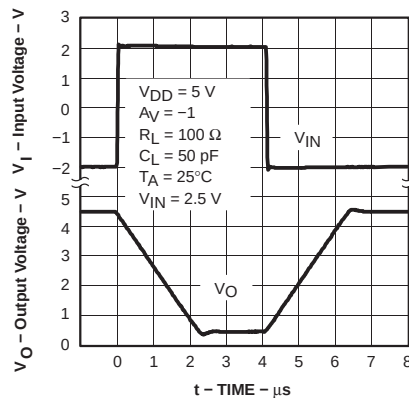


Figure 20

**INVERTING LARGE-SIGNAL
PULSE RESPONSE**

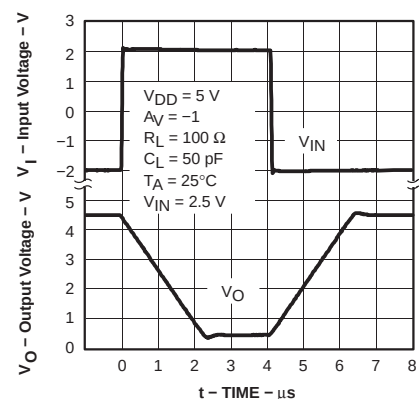


Figure 21

**SMALL-SIGNAL INVERTING
PULSE RESPONSE**

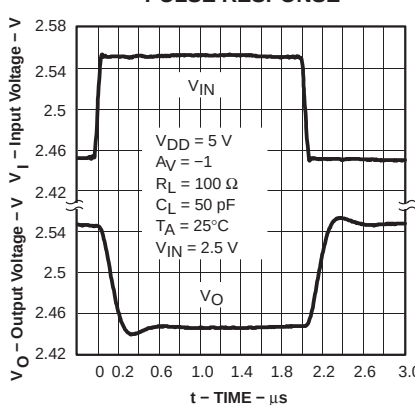


Figure 22

**CROSSTALK
VS
FREQUENCY**

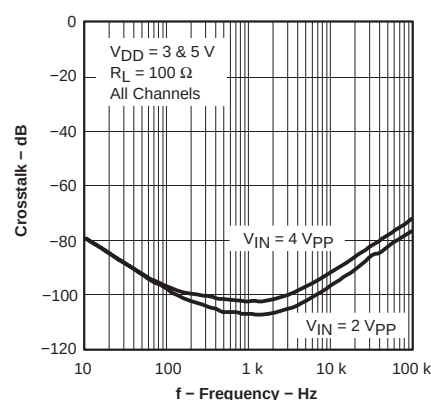


Figure 23

**SHUTDOWN FORWARD AND
REVERSE ISOLATION**

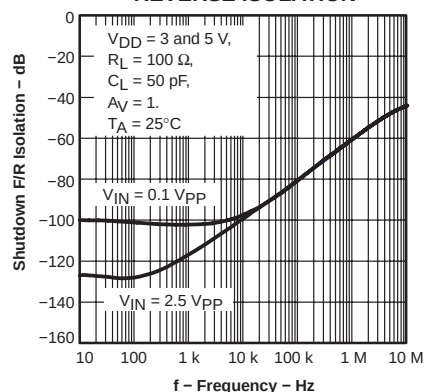


Figure 24

**SHUTDOWN SUPPLY CURRENT
VS
FREE-AIR TEMPERATURE**

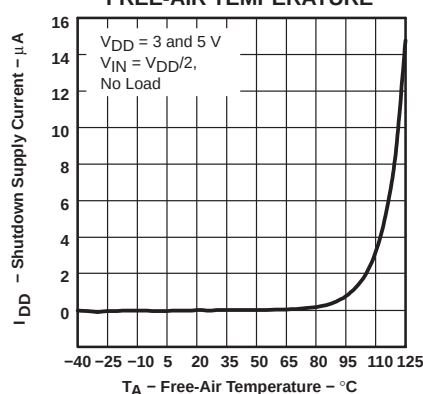


Figure 25

TYPICAL CHARACTERISTICS

SHUTDOWN SUPPLY CURRENT / OUTPUT VOLTAGE

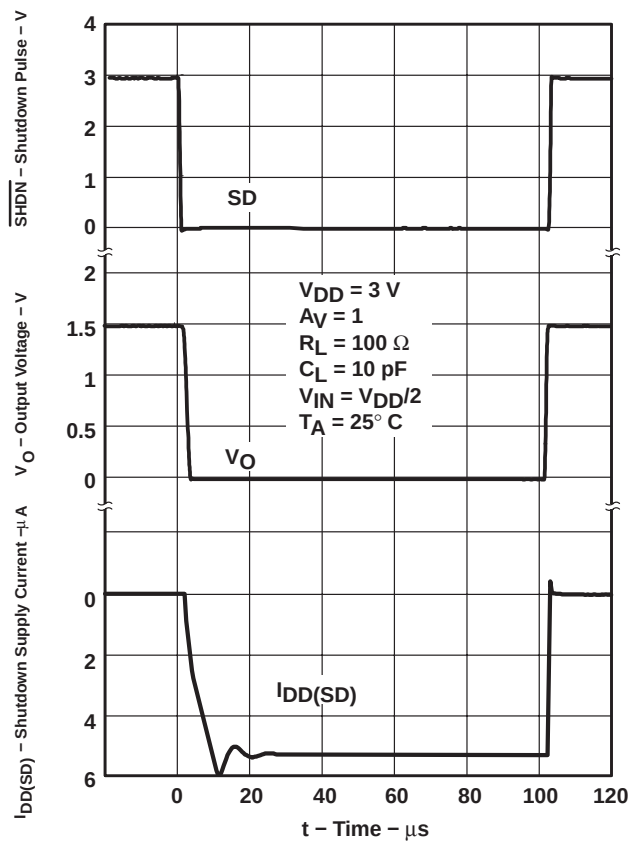


Figure 26

TLV4110, TLV4111, TLV4112, TLV4113

FAMILY OF HIGH OUTPUT DRIVE OPERATIONAL AMPLIFIERS WITH SHUTDOWN

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APPLICATION INFORMATION

shutdown function

Two members of the TLV411x family (TLV4110/3) have a shutdown terminal for conserving battery life in portable applications. When the shutdown terminal is tied low, the supply current is reduced to just nano amps per channel, the amplifier is disabled, and the outputs are placed in a high impedance mode. In order to save power in shutdown mode, an external pullup resistor is required, therefore, to enable the amplifier the shutdown terminal must be pulled high. When the shutdown terminal is left floating, care should be taken to ensure that parasitic leakage current at the shutdown terminal does not inadvertently place the operational amplifier into shutdown.

driving a capacitive load

When the amplifier is configured in this manner, capacitive loading directly on the output will decrease the device's phase margin leading to high frequency ringing or oscillations. Therefore, for capacitive loads of greater than 1 nF, it is recommended that a resistor be placed in series (R_{NULL}) with the output of the amplifier, as shown in Figure 27. A maximum value of 20 Ω should work well for most applications.

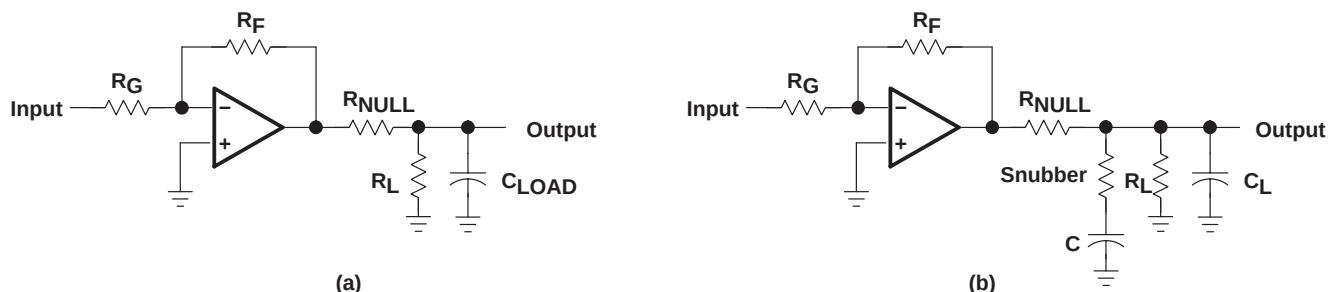


Figure 27. Driving a Capacitive Load

offset voltage

The output offset voltage, (V_{OO}) is the sum of the input offset voltage (V_{IO}) and both input bias currents (I_{IB}) times the corresponding gains. The following schematic and formula can be used to calculate the output offset voltage:

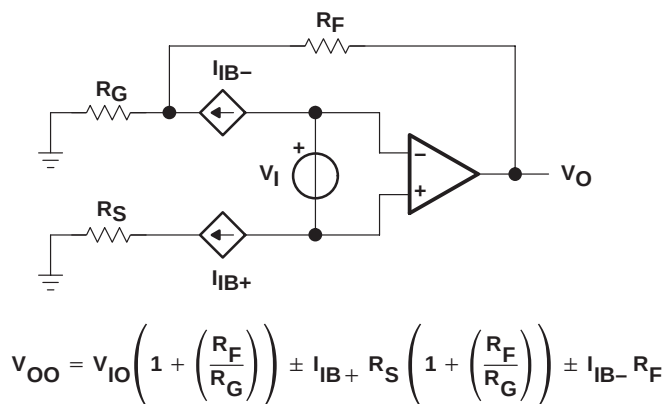


Figure 28. Output Offset Voltage Model

APPLICATION INFORMATION

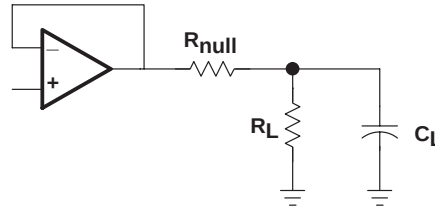


Figure 29

general power design considerations

When driving heavy loads at high junction temperatures there is an increased probability of electromigration affecting the long term reliability of ICs. Therefore for this not to be an issue either:

- The output current must be limited (at these high junction temperatures).
- or
- The junction temperature must be limited.

The maximum continuous output current at a die temperature 150°C will be 1/3 of the current at 105°C.

The junction temperature will be dependent on the ambient temperature around the IC, thermal impedance from the die to the ambient and power dissipated within the IC.

$$T_J = T_A + \theta_{JA} \times P_{DIS}$$

Where:

P_{DIS} is the IC power dissipation and is equal to the output current multiplied by the voltage dropped across the output of the IC.

θ_{JA} is the thermal impedance between the junction and the ambient temperature of the IC.

T_J is the junction temperature.

T_A is the ambient temperature.

Reducing one or more of these factors results in a reduced die temperature. The 8-pin SOIC (small outline integrated circuit) has a thermal impedance from junction to ambient of 176°C/W. For this reason it is recommended that the maximum power dissipation of the 8-pin SOIC package be limited to 350 mW, with peak dissipation of 700 mW as long as the RMS value is less than 350 mW.

The use of the MSOP PowerPAD™ dramatically reduces the thermal impedance from junction to case. And with correct mounting, the reduced thermal impedance greatly increases the IC's permissible power dissipation and output current handling capability. For example, the power dissipation of the PowerPAD™ is increased to above 1 W. Sinusoidal and pulse-width modulated output signals also increase the output current capability. The equivalent dc current is proportional to the square-root of the duty cycle:

$$I_{DC(EQ)} = I_{Cont} \times \sqrt{(\text{duty cycle})}$$

CURRENT DUTY CYCLE AT PEAK RATED CURRENT	EQUIVALENT DC CURRENT AS A PERCENTAGE OF PEAK
100	100
70	84
50	71

Note that with an operational amplifier, a duty cycle of 70% would often result in the op amp sourcing current 70% of the time and sinking current 30%, therefore, the equivalent dc current would still be 0.84 times the continuous current rating at a particular junction temperature.

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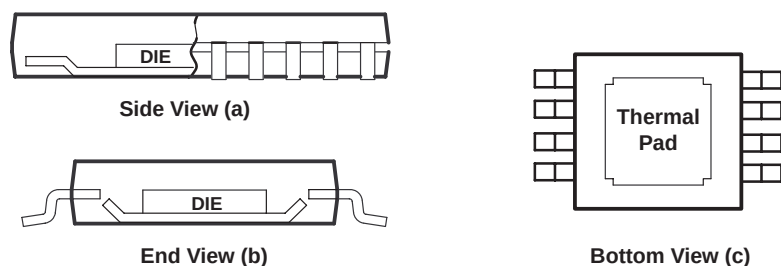
general PowerPAD design considerations

The TLV411x is available in a thermally-enhanced PowerPAD family of packages. These packages are constructed using a downset leadframe upon which the die is mounted [see Figure 30(a) and Figure 30(b)]. This arrangement results in the lead frame being exposed as a thermal pad on the underside of the package [see Figure 30(c)]. Because this thermal pad has direct thermal contact with the die, excellent thermal performance can be achieved by providing a good thermal path away from the thermal pad.

The PowerPAD package allows for both assembly and thermal management in one manufacturing operation. During the surface-mount solder operation (when the leads are being soldered), the thermal pad must be soldered to a copper area underneath the package. Through the use of thermal paths within this copper area, heat can be conducted away from the package into either a ground plane or other heat dissipating device.

Soldering the PowerPAD to the PCB is always recommended, even with applications that have low-power dissipation. This provides the necessary thermal and mechanical connection between the lead frame die pad and the PCB.

The PowerPAD package represents a breakthrough in combining the small area and ease of assembly of surface mount with mechanical methods of heatsinking.



NOTE A: The thermal pad is electrically isolated from all terminals in the package.

Figure 30. Views of Thermally-Enhanced DGN Package

APPLICATION INFORMATION

Although there are many ways to properly heatsink the PowerPAD package, the following steps illustrate the recommended approach.

general PowerPAD design considerations (continued)

1. The thermal pad must be connected to the most negative supply voltage on the device, GND.
2. Prepare the PCB with a top side etch pattern as illustrated in the thermal land pattern mechanical drawings at the end of this document. There should be etch for the leads as well as etch for the thermal pad.
3. Place five holes in the area of the thermal pad. These holes should be 13 mils in diameter. Keep them small so that solder wicking through the holes is not a problem during reflow.
4. Additional vias may be placed anywhere along the thermal plane outside of the thermal pad area. This helps dissipate the heat generated by the TLV411x IC. These additional vias may be larger than the 13-mil diameter vias directly under the thermal pad. They can be larger because they are not in the thermal pad area to be soldered so that wicking is not a problem.
5. Connect all holes to the internal ground plane that is at the same voltage potential as the device GND pin.
6. When connecting these holes to the ground plane, do not use the typical web or spoke via connection methodology. Web connections have a high thermal resistance connection that is useful for slowing the heat transfer during soldering operations. This makes the soldering of vias that have plane connections easier. In this application, however, low thermal resistance is desired for the most efficient heat transfer. Therefore, the holes under the TLV411x PowerPAD package should make their connection to the internal ground plane with a complete connection around the entire circumference of the plated-through hole.
7. The top-side solder mask should leave the terminals of the package and the thermal pad area with its five holes exposed. The bottom-side solder mask should cover the five holes of the thermal pad area. This prevents solder from being pulled away from the thermal pad area during the reflow process.
8. Apply solder paste to the exposed thermal pad area and all of the IC terminals.
9. With these preparatory steps in place, the TLV411x IC is simply placed in position and run through the solder reflow operation as any standard surface-mount component. This results in a part that is properly installed.

For a given θ_{JA} , the maximum power dissipation is shown in Figure 31 and is calculated by the following formula:

$$P_D = \left(\frac{T_{MAX} - T_A}{\theta_{JA}} \right)$$

Where:

P_D = Maximum power dissipation of TLV411x IC (watts)

T_{MAX} = Absolute maximum junction temperature (150°C)

T_A = Free-ambient air temperature (°C)

θ_{JA} = $\theta_{JC} + \theta_{CA}$

θ_{JC} = Thermal coefficient from junction to case

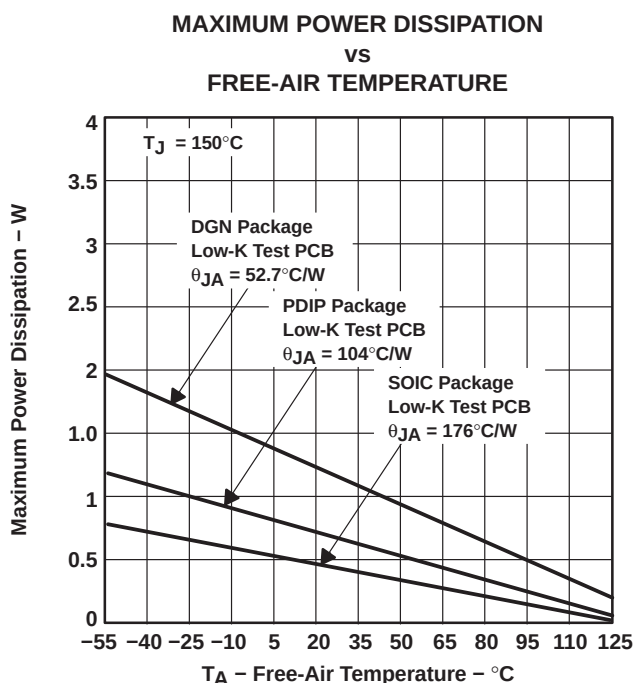
θ_{CA} = Thermal coefficient from case to ambient air (°C/W)

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APPLICATION INFORMATION

general PowerPAD design considerations (continued)



NOTE A: Results are with no air flow and using JEDEC Standard Low-K test PCB.

Figure 31. Maximum Power Dissipation vs Free-Air Temperature

The next consideration is the package constraints. The two sources of heat within an amplifier are quiescent power and output power. The designer should never forget about the quiescent heat generated within the device, especially multi-amplifier devices. Because these devices have linear output stages (Class A-B), most of the heat dissipation is at low output voltages with high output currents.

The other key factor when dealing with power dissipation is how the devices are mounted on the PCB. The PowerPAD devices are extremely useful for heat dissipation. But, the device should always be soldered to a copper plane to fully use the heat dissipation properties of the PowerPAD. The SOIC package, on the other hand, is highly dependent on how it is mounted on the PCB. As more trace and copper area is placed around the device, θ_{JA} decreases and the heat dissipation capability increases. The currents and voltages shown in these graphs are for the total package. For the dual amplifier packages, the sum of the RMS output currents and voltages should be used to choose the proper package.

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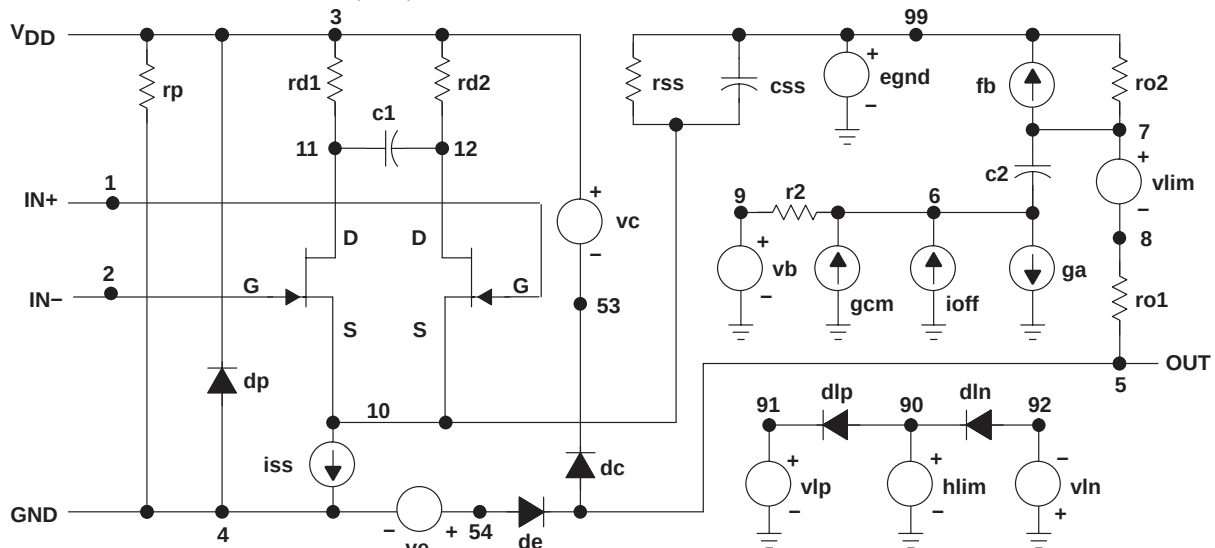
APPLICATION INFORMATION

macromodel information

Macromodel information provided was derived using Microsim *Parts*™, the model generation software used with Microsim *PSpice*™. The Boyle macromodel (see Note 3) and subcircuit in Figure 33 are generated using the TLV411x typical electrical and operating characteristics at $T_A = 25^\circ\text{C}$. Using this information, output simulations of the following key parameters can be generated to a tolerance of 20% (in most cases):

- Maximum positive output voltage swing
- Maximum negative output voltage swing
- Slew rate
- Quiescent power dissipation
- Input bias current
- Open-loop voltage amplification
- Unity-gain frequency
- Common-mode rejection ratio
- Phase margin
- DC output resistance
- AC output resistance
- Short-circuit output current limit

NOTE 3: G. R. Boyle, B. M. Cohn, D. O. Pederson, and J. E. Solomon, "Macromodeling of Integrated Circuit Operational Amplifiers," *IEEE Journal of Solid-State Circuits*, SC-9, 353 (1974).



* TLV4112_5V operational amplifier "macromodel" subcircuit
* updated using Model Editor release 9.1 on 01/18/00 at 15:50
Model Editor is an OrCAD product.

* connections: non-inverting input
* | inverting input
* | positive power supply
* | negative power supply
* | output

```
.subckt TLV4112_5V 1 2 3 4 5
*
c1      11      12      2.2439E-12
c2      6       7       10.000E-12
css     5       99      454.55E-15
dc      5       53      dy
de      54      5       dy
dlp     90      91      dx
dln     92      90      dx
dp      4       3       dx
egnd    99      0       poly(2) (3,0) (4,0) 0 .5 .5
fb      7       99      poly(5) vb vc ve vlp vln 0
+ 33.395E6 -1E3 1E3 33E6 -33E6
ga      6       0       11      12 168.39E-6
gcm     0       6       10      99 168.39E-12
```

```
iss      10      4      dc      13.800E-6
hlim     90      0      vlim 1K
ioff     0       6      dc      75E-9
j1       11      2      10 jx1
j2       12      1      10 jx2
r2       6       9      100.00E3
rd1      3       11     5.9386E3
rd2      3       12     5.9386E3
ro1      8       5      10
ro2      7       99     10
rp       3       4      3.3333E3
rss     10      99     14.493E6
vb       9       0      dc 0
vc       3       53     dc .86795
ve      54      4      dc .86795
vlim     7       8      dc 0
vlp     91      0      dc 300
vln     0       92     dc 300
.model   dx      D(Is=800.00E-18)
.model   dy      D(Is=800.00E-18 Rs=1m Cjo=10p)
.model   jx1     NJF(Is=150.00E-12 Beta=2.0547E-3 +Vto=-1)
.model   jx2     NJF(Is=150.00E-12 Beta=2.0547E-3 +Vto=-1)
.ends
*$
```

Figure 32. Boyle Macromodel and Subcircuit

PSpice and Parts are trademarks of MicroSim Corporation.



PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
TLV4110ID	Active	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4110I
TLV4110ID.A	Active	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4110I
TLV4110IDGNR	Active	Production	HVSSOP (DGN) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	AHM
TLV4110IDGNR.A	Active	Production	HVSSOP (DGN) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	AHM
TLV4110IDR	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4110I
TLV4110IDR.A	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4110I
TLV4110IP	Active	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 125	TLV4110I
TLV4110IP.A	Active	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 125	TLV4110I
TLV4111CD	Active	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	4111C
TLV4111CD.A	Active	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	4111C
TLV4111CDGN	Active	Production	HVSSOP (DGN) 8	80 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	AHN
TLV4111CDGN.A	Active	Production	HVSSOP (DGN) 8	80 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	AHN
TLV4111ID	Active	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4111I
TLV4111ID.A	Active	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4111I
TLV4111IDGN	Active	Production	HVSSOP (DGN) 8	80 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	AHO
TLV4111IDGN.A	Active	Production	HVSSOP (DGN) 8	80 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	AHO
TLV4111IDGNR	Active	Production	HVSSOP (DGN) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	AHO
TLV4111IDGNR.A	Active	Production	HVSSOP (DGN) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	AHO
TLV4111IDR	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4111I
TLV4111IDR.A	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4111I
TLV4112CD	Active	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	4112C
TLV4112CD.A	Active	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	4112C
TLV4112CDGN	Active	Production	HVSSOP (DGN) 8	80 TUBE	Yes	NIPDAU NIPDAUAG	Level-1-260C-UNLIM	0 to 70	AHP
TLV4112CDGN.A	Active	Production	HVSSOP (DGN) 8	80 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	AHP
TLV4112CP	Active	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	TLV4112C
TLV4112CP.A	Active	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	TLV4112C
TLV4112ID	Active	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4112I
TLV4112ID.A	Active	Production	SOIC (D) 8	75 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4112I
TLV4112IDGN	Active	Production	HVSSOP (DGN) 8	80 TUBE	Yes	NIPDAU NIPDAUAG	Level-1-260C-UNLIM	-40 to 125	AHQ

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
TLV4112IDGN.A	Active	Production	HVSSOP (DGN) 8	80 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	AHQ
TLV4112IDGNR	Active	Production	HVSSOP (DGN) 8	2500 LARGE T&R	Yes	NIPDAU NIPDAUAG	Level-1-260C-UNLIM	-40 to 125	AHQ
TLV4112IDGNR.A	Active	Production	HVSSOP (DGN) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	AHQ
TLV4112IDR	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4112I
TLV4112IDR.A	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4112I
TLV4112IP	Active	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 125	TLV4112I
TLV4112IP.A	Active	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 125	TLV4112I
TLV4113ID	Active	Production	SOIC (D) 14	50 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4113I
TLV4113ID.A	Active	Production	SOIC (D) 14	50 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	4113I
TLV4113IDGQ	Active	Production	HVSSOP (DGQ) 10	80 TUBE	Yes	Call TI Nipdau	Level-1-260C-UNLIM	-40 to 125	AHS
TLV4113IDGQ.A	Active	Production	HVSSOP (DGQ) 10	80 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	AHS
TLV4113IDGQR	Active	Production	HVSSOP (DGQ) 10	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	AHS
TLV4113IDGQR.A	Active	Production	HVSSOP (DGQ) 10	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	AHS
TLV4113IN	Active	Production	PDIP (N) 14	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 125	TLV4113I
TLV4113IN.A	Active	Production	PDIP (N) 14	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 125	TLV4113I

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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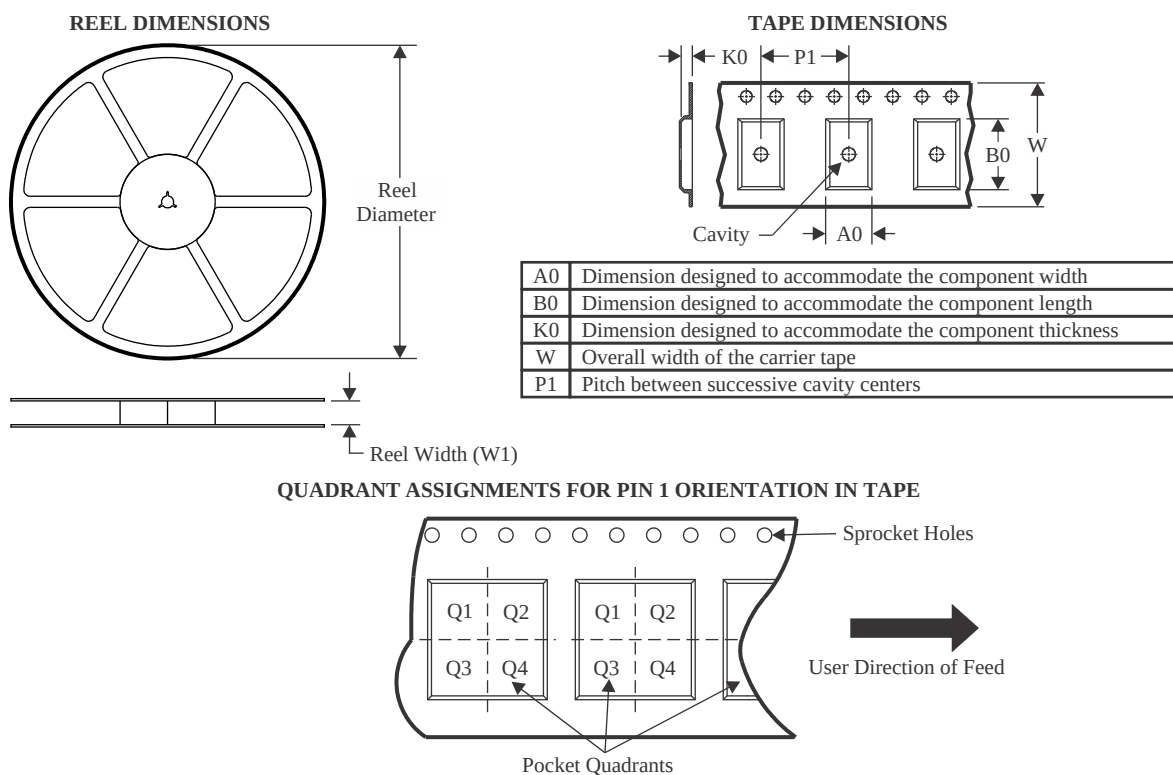
OTHER QUALIFIED VERSIONS OF TLV4113 :

- Enhanced Product : [TLV4113-EP](#)

NOTE: Qualified Version Definitions:

- Enhanced Product - Supports Defense, Aerospace and Medical Applications

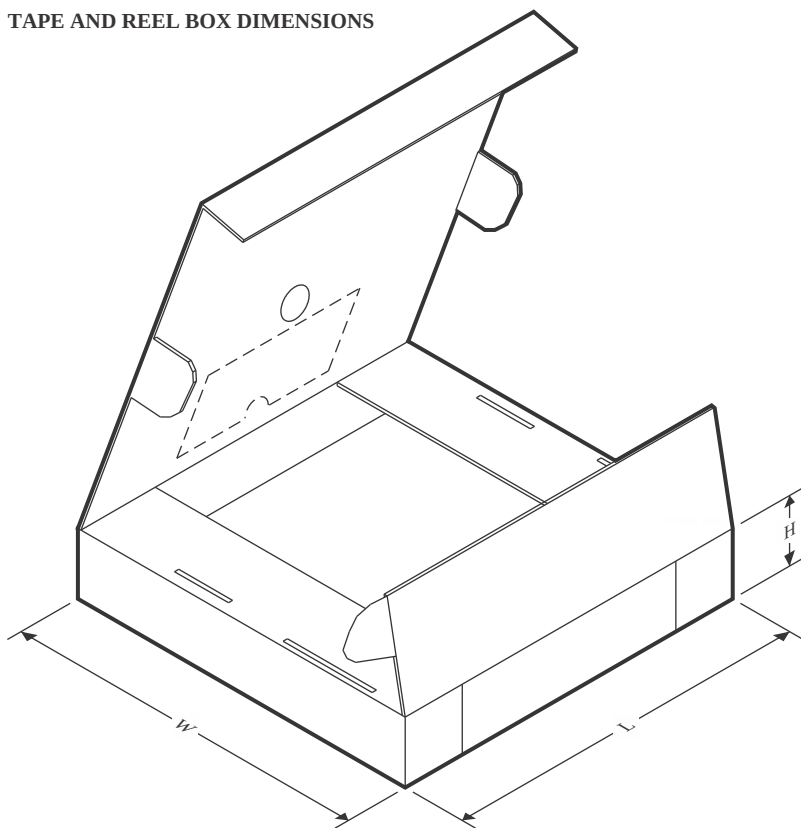
TAPE AND REEL INFORMATION



*All dimensions are nominal

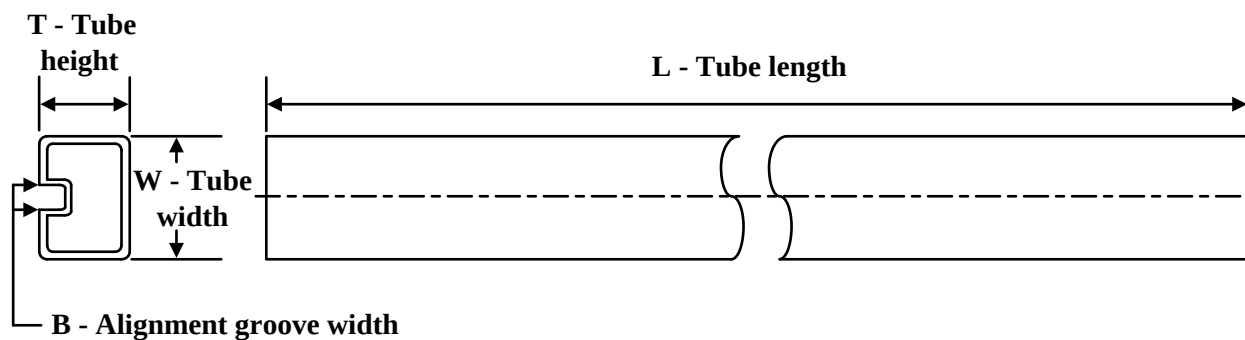
Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
TLV4110IDGNR	HVSSOP	DGN	8	2500	330.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1
TLV4110IDR	SOIC	D	8	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
TLV4111IDGNR	HVSSOP	DGN	8	2500	330.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1
TLV4111IDR	SOIC	D	8	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
TLV4112IDGNR	HVSSOP	DGN	8	2500	330.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1
TLV4112IDR	SOIC	D	8	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
TLV4113IDGQR	HVSSOP	DGQ	10	2500	330.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1

TAPE AND REEL BOX DIMENSIONS



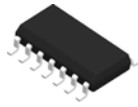
*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
TLV4110IDGNR	HVSSOP	DGN	8	2500	358.0	335.0	35.0
TLV4110IDR	SOIC	D	8	2500	353.0	353.0	32.0
TLV4111IDGNR	HVSSOP	DGN	8	2500	358.0	335.0	35.0
TLV4111IDR	SOIC	D	8	2500	353.0	353.0	32.0
TLV4112IDGNR	HVSSOP	DGN	8	2500	358.0	335.0	35.0
TLV4112IDR	SOIC	D	8	2500	353.0	353.0	32.0
TLV4113IDGQR	HVSSOP	DGQ	10	2500	358.0	335.0	35.0

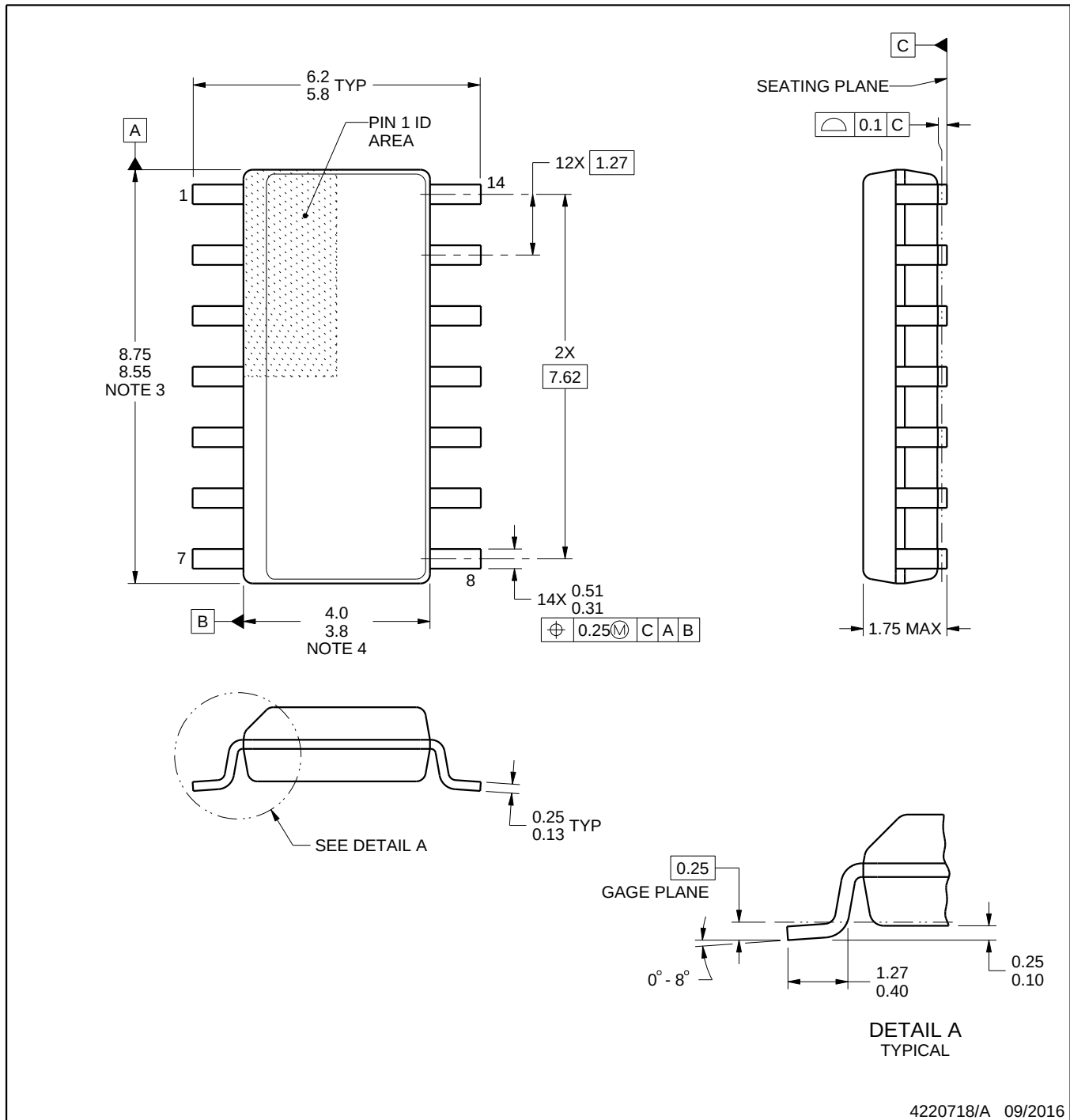
TUBE


*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
TLV4110ID	D	SOIC	8	75	507	8	3940	4.32
TLV4110ID.A	D	SOIC	8	75	507	8	3940	4.32
TLV4110IP	P	PDIP	8	50	506	13.97	11230	4.32
TLV4110IP.A	P	PDIP	8	50	506	13.97	11230	4.32
TLV4111CD	D	SOIC	8	75	507	8	3940	4.32
TLV4111CD.A	D	SOIC	8	75	507	8	3940	4.32
TLV4111ID	D	SOIC	8	75	507	8	3940	4.32
TLV4111ID.A	D	SOIC	8	75	507	8	3940	4.32
TLV4112CD	D	SOIC	8	75	507	8	3940	4.32
TLV4112CD	D	SOIC	8	75	505.46	6.76	3810	4
TLV4112CD.A	D	SOIC	8	75	505.46	6.76	3810	4
TLV4112CD.A	D	SOIC	8	75	507	8	3940	4.32
TLV4112CDGN	DGN	HVSSOP	8	80	330	6.55	500	2.88
TLV4112CDGN.A	DGN	HVSSOP	8	80	330	6.55	500	2.88
TLV4112CP	P	PDIP	8	50	506	13.97	11230	4.32
TLV4112CP.A	P	PDIP	8	50	506	13.97	11230	4.32
TLV4112ID	D	SOIC	8	75	505.46	6.76	3810	4
TLV4112ID	D	SOIC	8	75	507	8	3940	4.32
TLV4112ID.A	D	SOIC	8	75	505.46	6.76	3810	4
TLV4112ID.A	D	SOIC	8	75	507	8	3940	4.32
TLV4112IDGN	DGN	HVSSOP	8	80	330	6.55	500	2.88
TLV4112IDGN.A	DGN	HVSSOP	8	80	330	6.55	500	2.88
TLV4112IP	P	PDIP	8	50	506	13.97	11230	4.32
TLV4112IP.A	P	PDIP	8	50	506	13.97	11230	4.32
TLV4113ID	D	SOIC	14	50	507	8	3940	4.32
TLV4113ID.A	D	SOIC	14	50	507	8	3940	4.32
TLV4113IN	N	PDIP	14	25	506	13.97	11230	4.32
TLV4113IN.A	N	PDIP	14	25	506	13.97	11230	4.32

D0014A**PACKAGE OUTLINE****SOIC - 1.75 mm max height**

SMALL OUTLINE INTEGRATED CIRCUIT

**NOTES:**

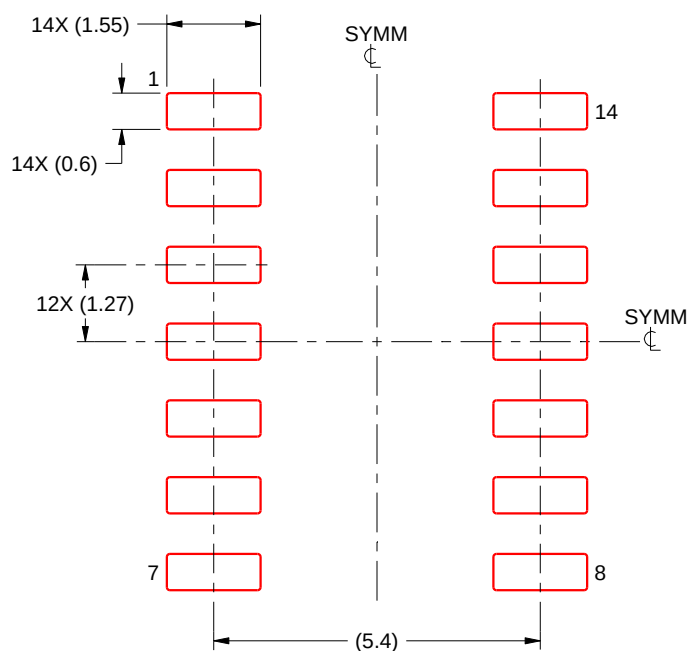
1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm, per side.
5. Reference JEDEC registration MS-012, variation AB.

EXAMPLE STENCIL DESIGN

D0014A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:8X

4220718/A 09/2016

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

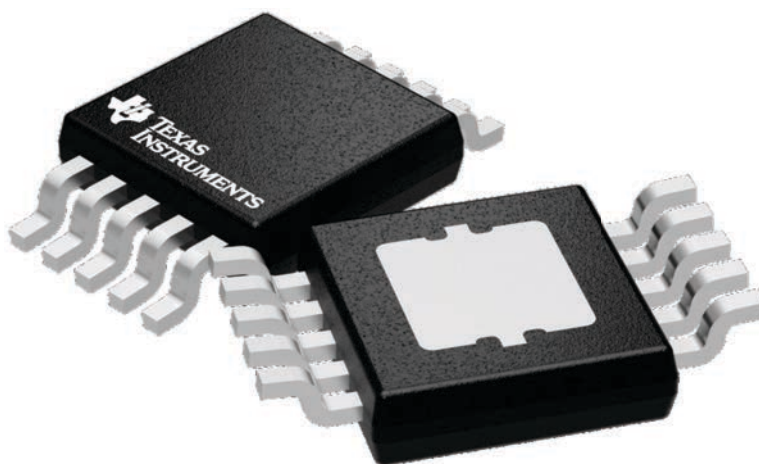
GENERIC PACKAGE VIEW

DGQ 10

PowerPAD™ HVSSOP - 1.1 mm max height

3 x 3, 0.5 mm pitch

PLASTIC SMALL OUTLINE



Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

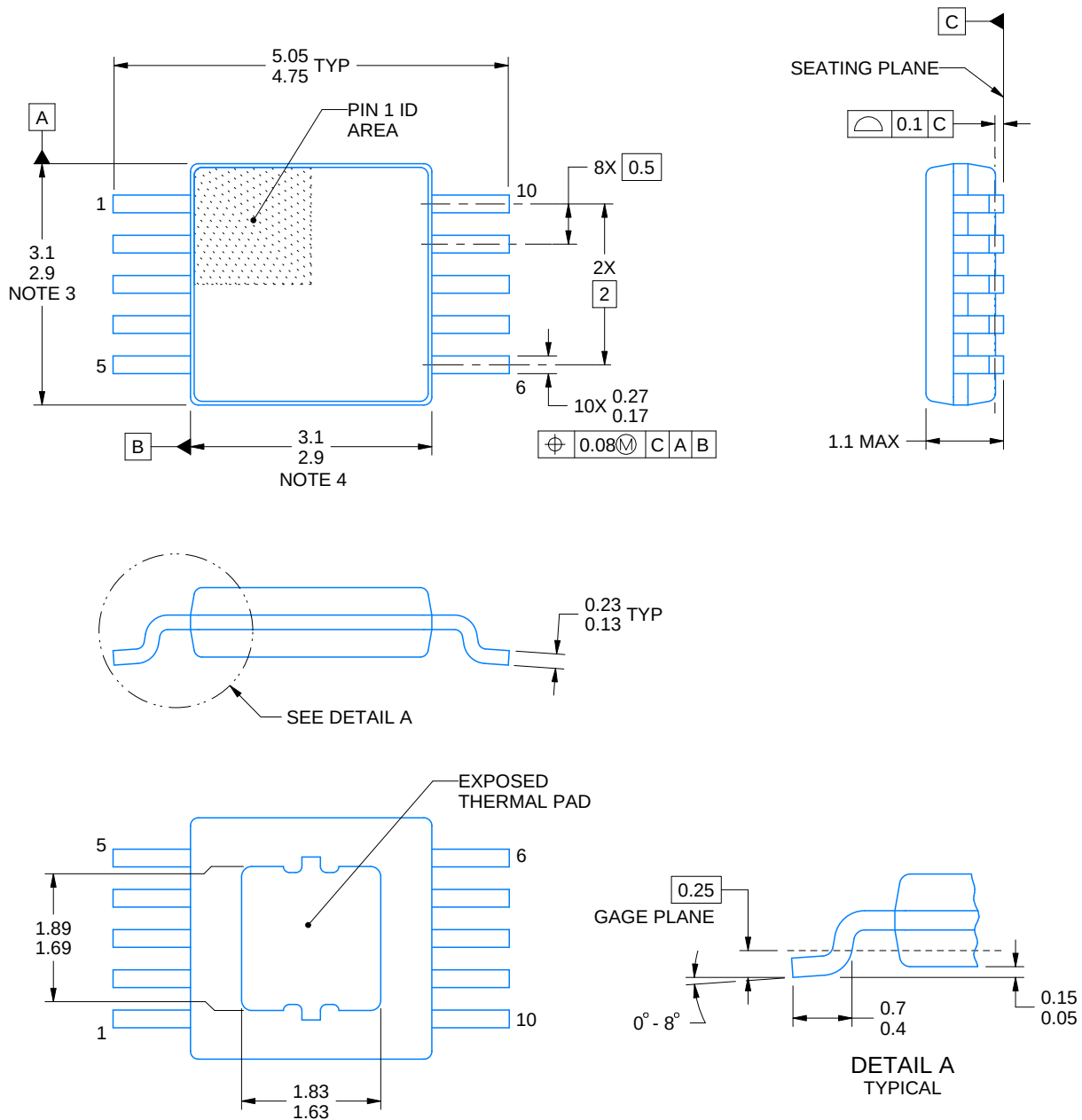
4224775/A

DGQ0010D

PACKAGE OUTLINE

PowerPAD™ - 1.1 mm max height

PLASTIC SMALL OUTLINE



4218842/B 04/2024

PowerPAD is a trademark of Texas Instruments.

NOTES:

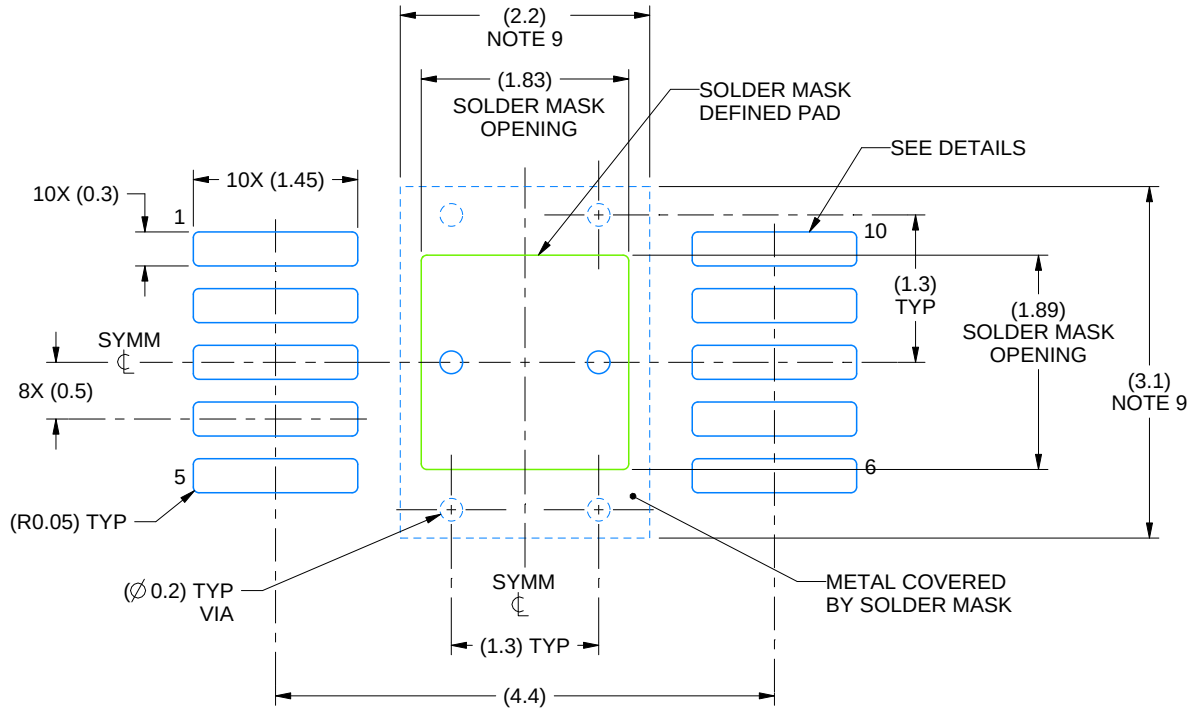
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-187, variation BA-T.

EXAMPLE BOARD LAYOUT

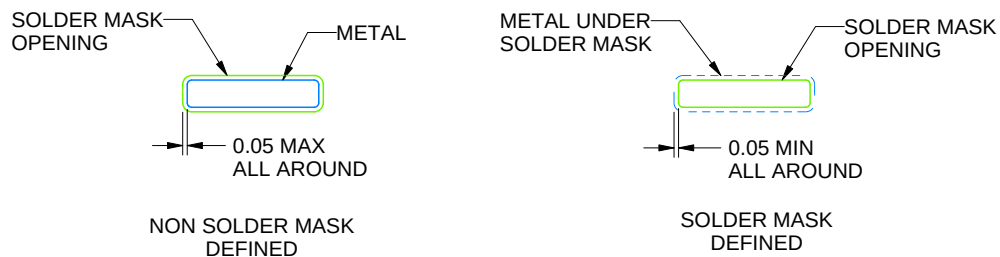
DGQ0010D

PowerPAD™ - 1.1 mm max height

PLASTIC SMALL OUTLINE



LAND PATTERN EXAMPLE
SCALE:15X



SOLDER MASK DETAILS

4218842/B 04/2024

NOTES: (continued)

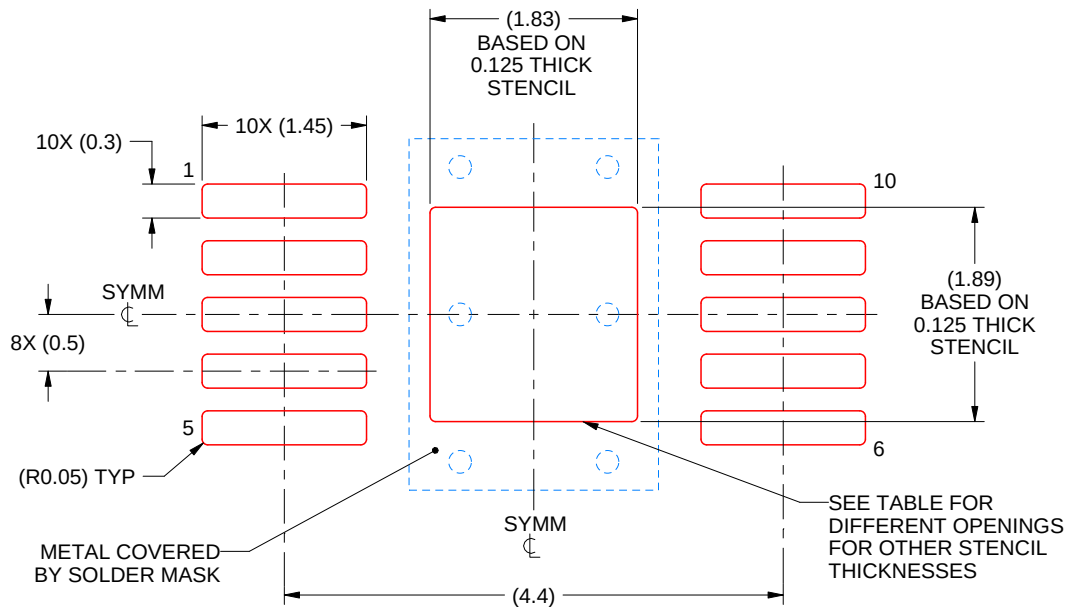
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
8. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature numbers SLMA002 (www.ti.com/lit/slma002) and SLMA004 (www.ti.com/lit/slma004).
9. Size of metal pad may vary due to creepage requirement.

EXAMPLE STENCIL DESIGN

DGQ0010D

PowerPAD™ - 1.1 mm max height

PLASTIC SMALL OUTLINE



SOLDER PASTE EXAMPLE
EXPOSED PAD
100% PRINTED SOLDER COVERAGE BY AREA
SCALE:15X

STENCIL THICKNESS	SOLDER STENCIL OPENING
0.1	2.05 X 2.11
0.125	1.83 X 1.89 (SHOWN)
0.150	1.67 X 1.73
0.175	1.55 X 1.60

4218842/B 04/2024

NOTES: (continued)

10. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
11. Board assembly site may have different recommendations for stencil design.

GENERIC PACKAGE VIEW

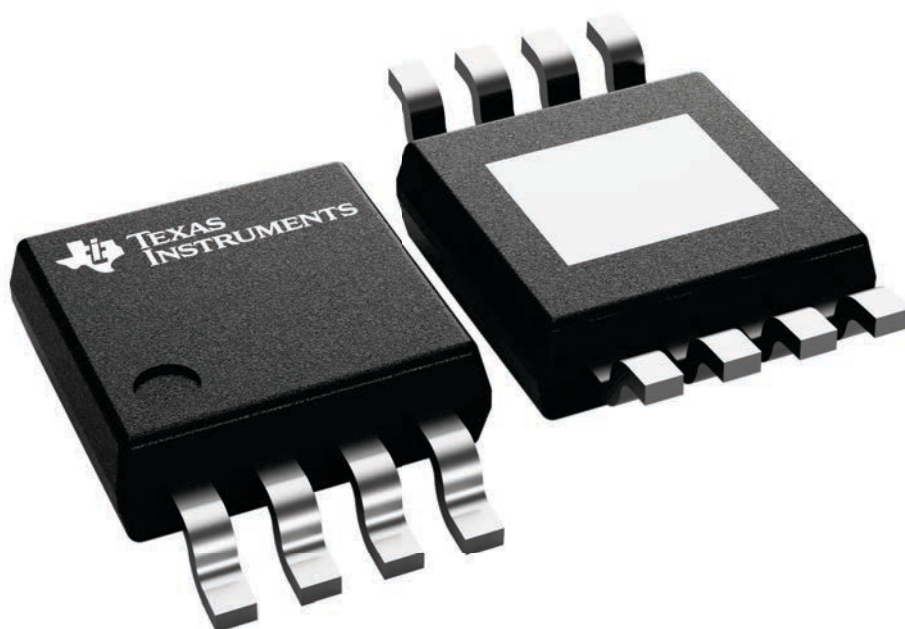
DGN 8

PowerPAD™ HVSSOP - 1.1 mm max height

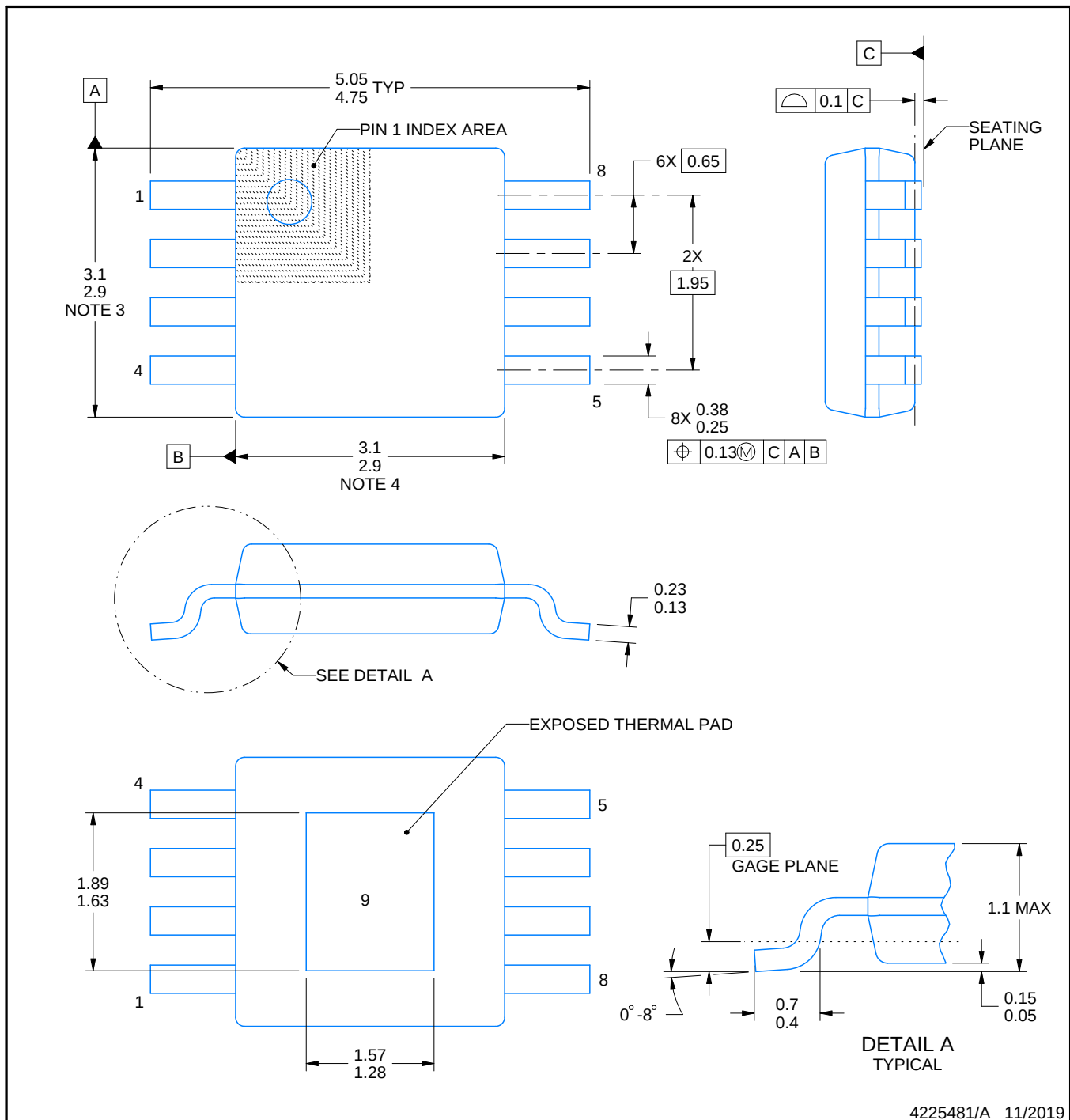
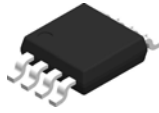
3 x 3, 0.65 mm pitch

SMALL OUTLINE PACKAGE

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.



4225482/B



4225481/A 11/2019

NOTES:

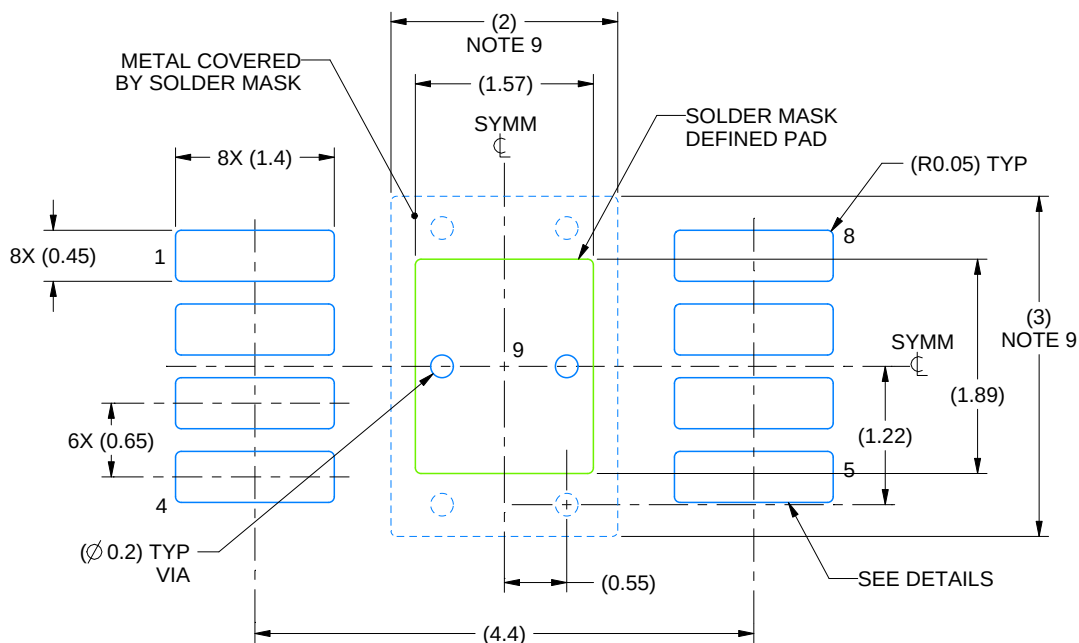
PowerPAD is a trademark of Texas Instruments.

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-187.

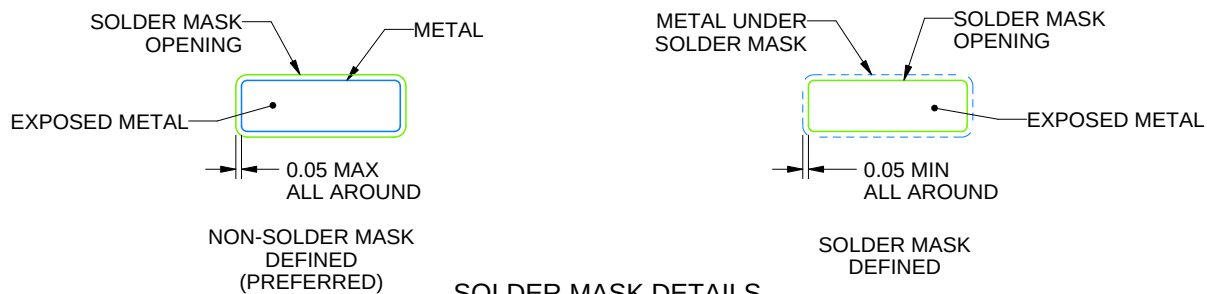
DGN0008D

PowerPAD™ VSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 15X



SOLDER MASK DETAILS

4225481/A 11/2019

NOTES: (continued)

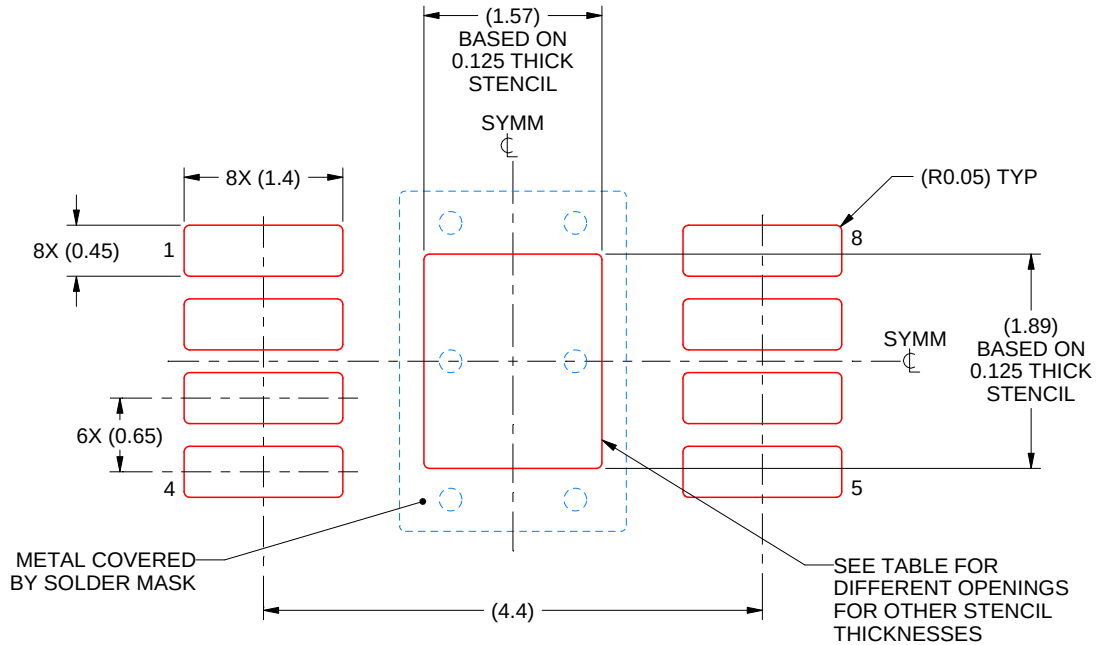
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
8. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.
9. Size of metal pad may vary due to creepage requirement.

EXAMPLE STENCIL DESIGN

DGN0008D

PowerPAD™ VSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



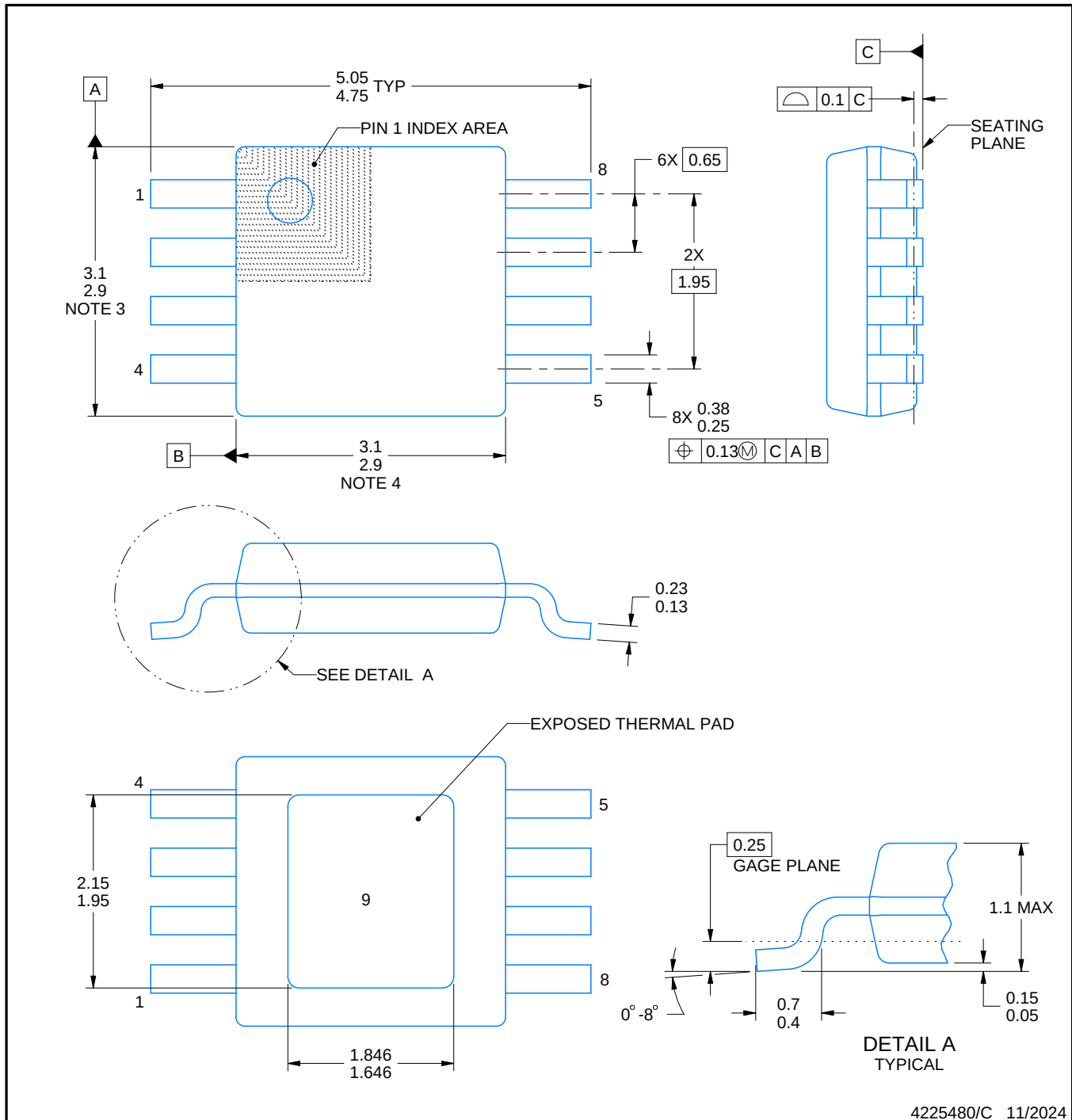
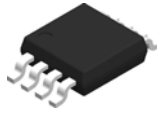
SOLDER PASTE EXAMPLE
EXPOSED PAD 9:
100% PRINTED SOLDER COVERAGE BY AREA
SCALE: 15X

STENCIL THICKNESS	SOLDER STENCIL OPENING
0.1	1.76 X 2.11
0.125	1.57 X 1.89 (SHOWN)
0.15	1.43 X 1.73
0.175	1.33 X 1.60

4225481/A 11/2019

NOTES: (continued)

10. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
11. Board assembly site may have different recommendations for stencil design.



NOTES:

PowerPAD is a trademark of Texas Instruments.

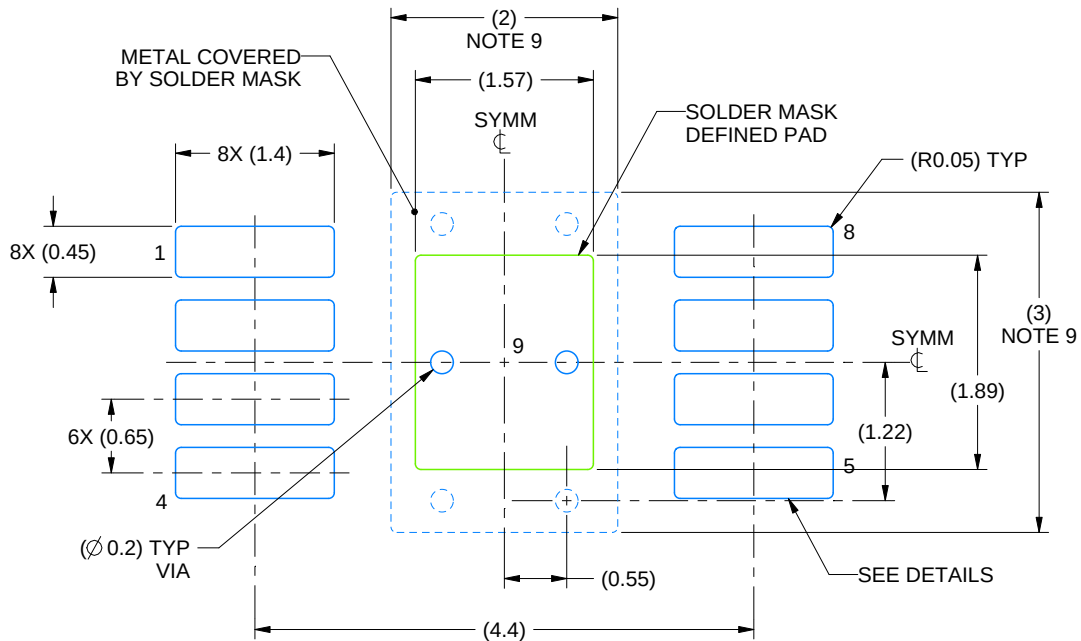
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-187.

EXAMPLE BOARD LAYOUT

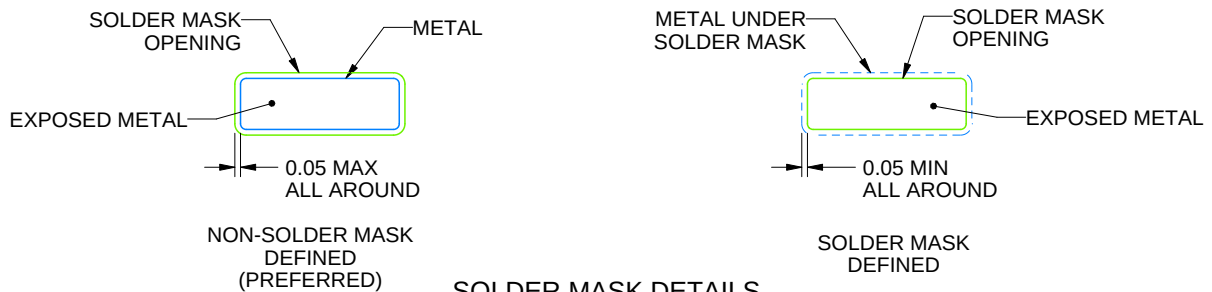
DGN0008G

PowerPAD™ HVSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 15X



SOLDER MASK DETAILS

4225480/C 11/2024

NOTES: (continued)

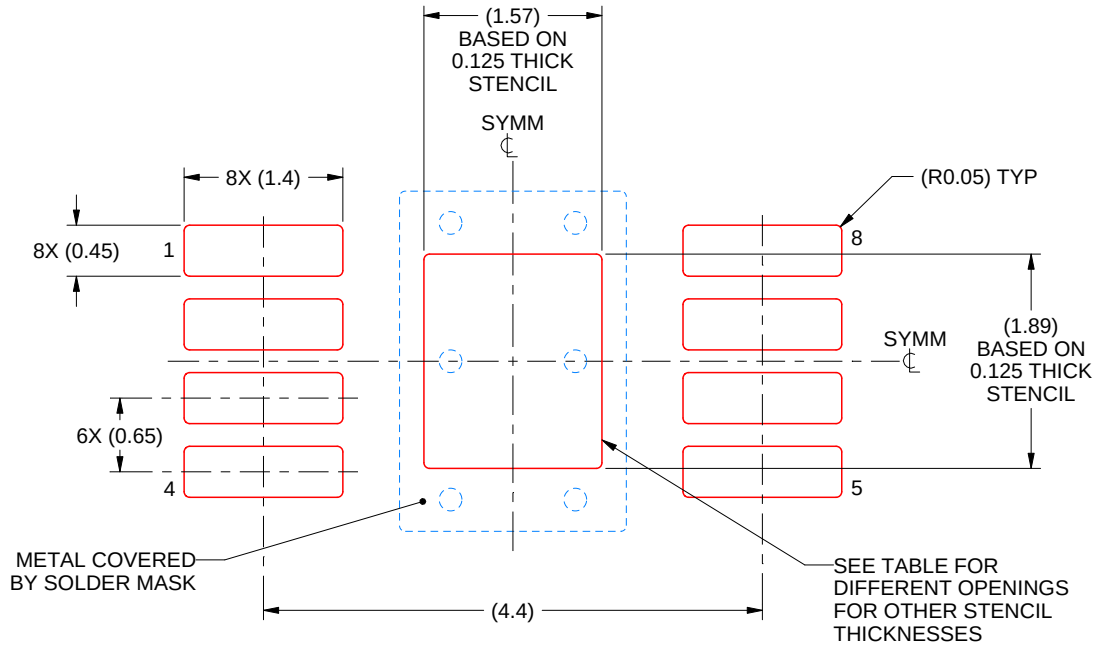
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
8. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.
9. Size of metal pad may vary due to creepage requirement.

EXAMPLE STENCIL DESIGN

DGN0008G

PowerPAD™ HVSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



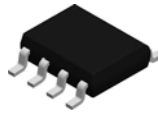
SOLDER PASTE EXAMPLE
EXPOSED PAD 9:
100% PRINTED SOLDER COVERAGE BY AREA
SCALE: 15X

STENCIL THICKNESS	SOLDER STENCIL OPENING
0.1	1.76 X 2.11
0.125	1.57 X 1.89 (SHOWN)
0.15	1.43 X 1.73
0.175	1.33 X 1.60

4225480/C 11/2024

NOTES: (continued)

10. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
11. Board assembly site may have different recommendations for stencil design.

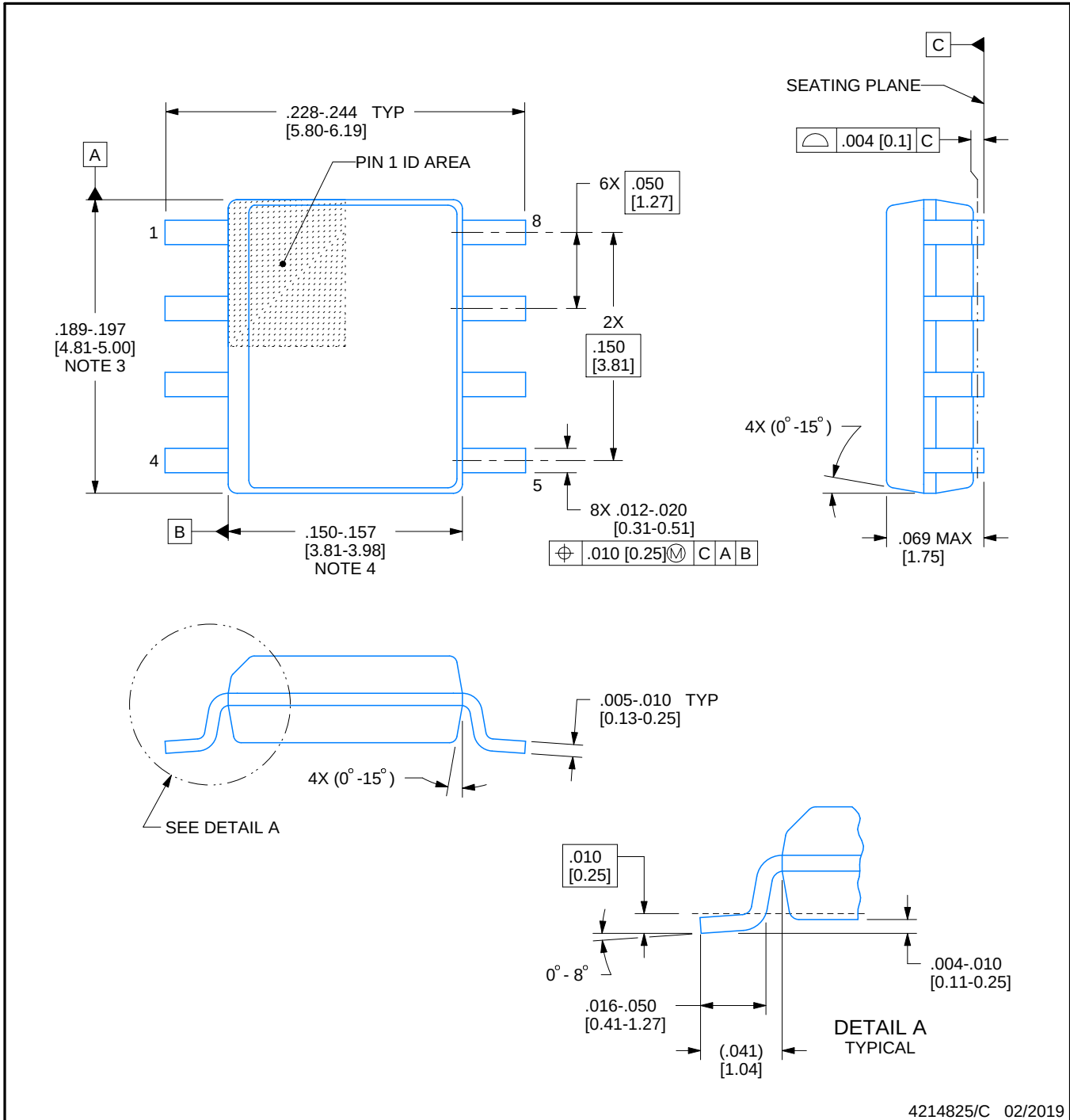


D0008A

PACKAGE OUTLINE

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



4214825/C 02/2019

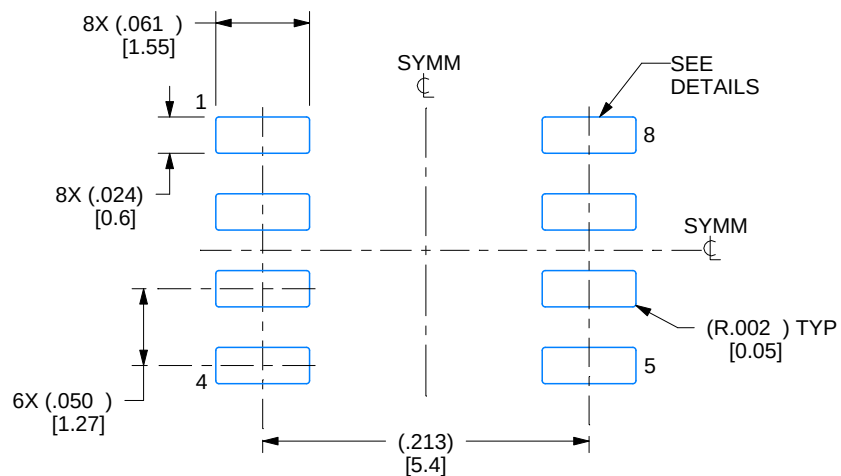
NOTES:

1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.

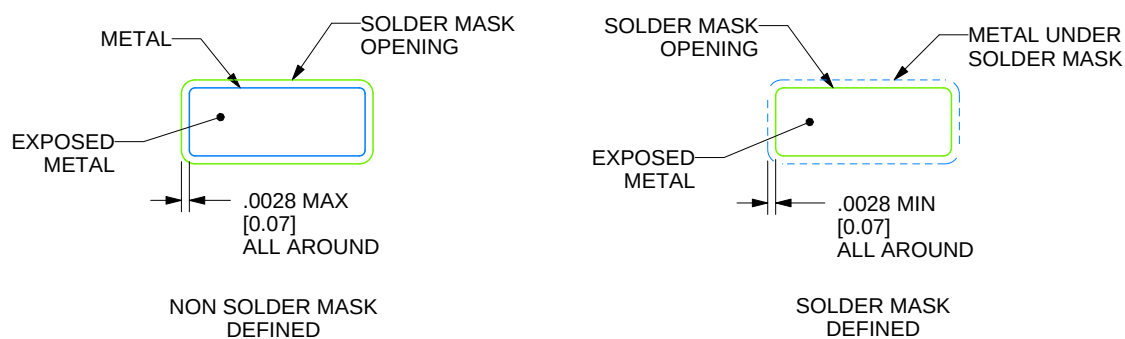
D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:8X



SOLDER MASK DETAILS

4214825/C 02/2019

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

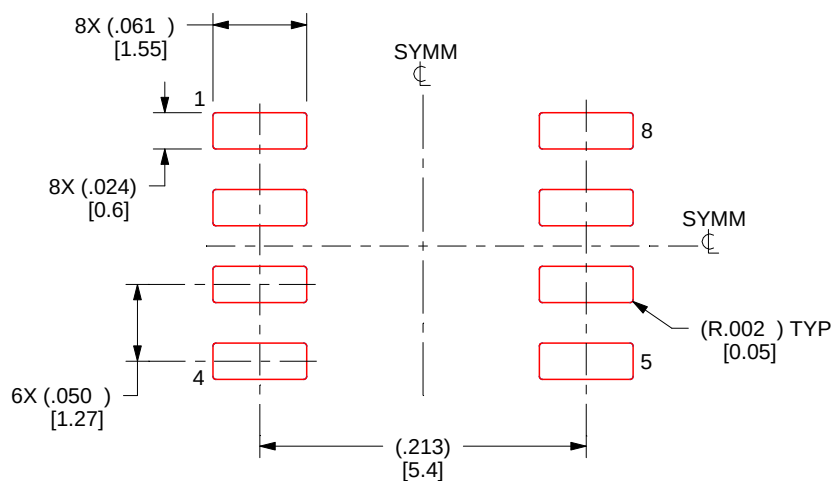
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE
BASED ON .005 INCH [0.125 MM] THICK STENCIL
SCALE:8X

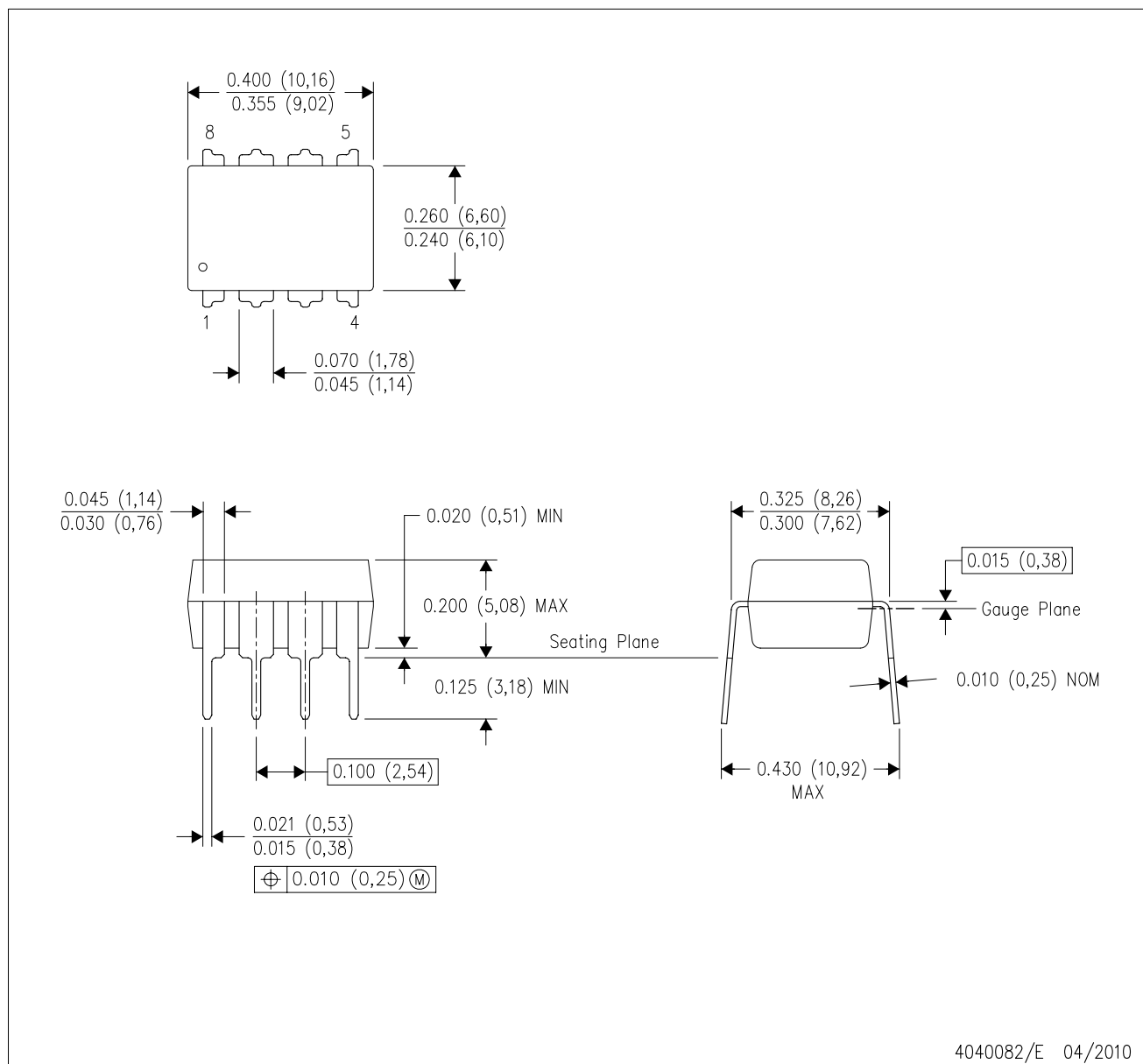
4214825/C 02/2019

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

P (R-PDIP-T8)

PLASTIC DUAL-IN-LINE PACKAGE

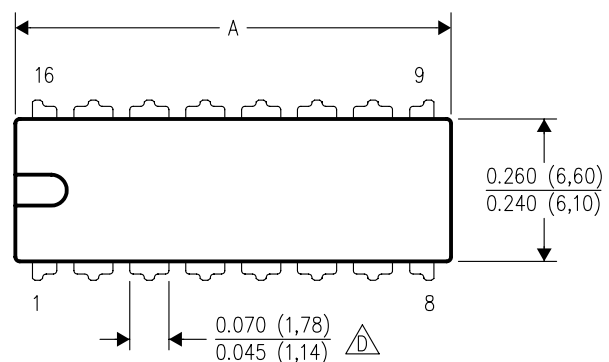


- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - C. Falls within JEDEC MS-001 variation BA.

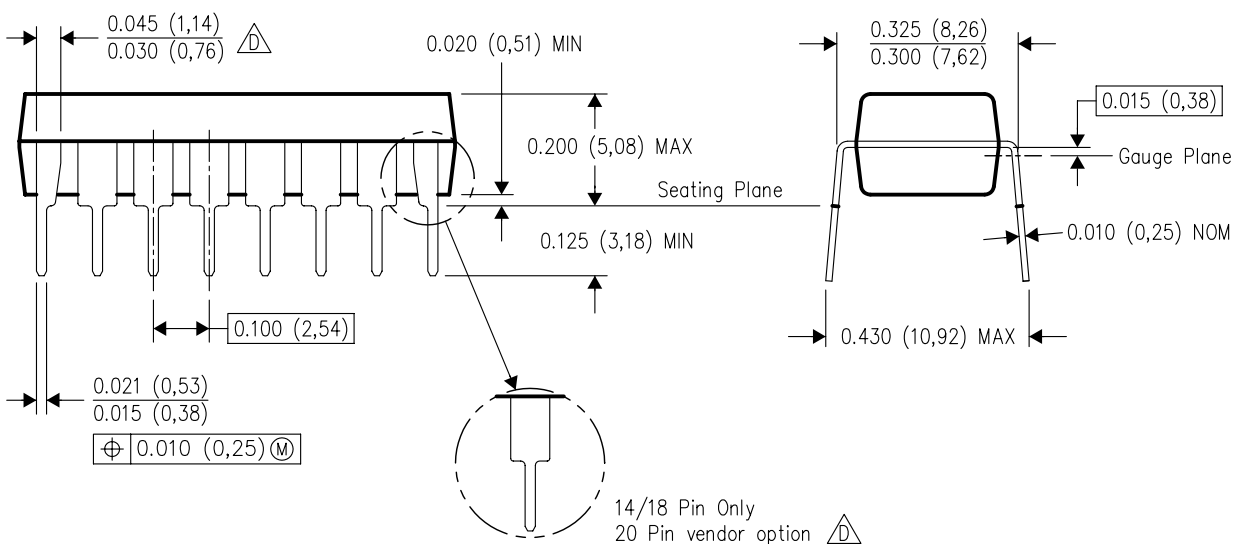
N (R-PDIP-T**)

16 PINS SHOWN

PLASTIC DUAL-IN-LINE PACKAGE



PINS ** DIM	14	16	18	20
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS-001 VARIATION	AA	BB	AC	AD



4040049/E 12/2002

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 -  Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).
 -  The 20 pin end lead shoulder width is a vendor option, either half or full width.

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